

Key Questions for Appraisal

NOTE: Often research vignettes that are followed by a set of questions in the exam contain some MCQs that require an opinion about the methodological rigour of a study. The following tips will help in answering these questions.

Key methodological points to consider in the appraisal of a RCT

Was the process of treatment allocation truly random?
Would participants have been able to know or guess their treatment allocation?
Were participants and researchers 'blinded' to participants' treatment group?
Were outcomes assessed objectively?
Were all participants who were randomly allocated a treatment accounted for in the final analysis?
Were all participants' data analyzed in the group to which they were randomly allocated?

Key methodological points to consider in the appraisal of systematic reviews and meta-analyses.

Were all relevant studies included (i.e. was the search comprehensive, did it exclude articles on the basis of publication status or language and was the potential for publication bias assessed)?
Were selected articles appraised and data extracted by two independent reviewers?
Was sufficient detail provided about the primary studies, including descriptions of the patients, interventions and outcomes?
Was the quality of the primary studies assessed?
Did the researchers assess the appropriateness of combining results to calculate a summary measure?

Key methodological points to consider in the appraisal of a cohort study.

Is the study prospective or retrospective?
Is the cohort representative of a defined group or population?
Were all important confounding factors identified?
Were all important exposures and/or treatments, potential confounding factors and outcomes measured accurately and objectively in all members of the cohort?
Were there important losses to follow-up?
Were participants followed up for a sufficient length of time?

Key methodological points to consider in the appraisal of a case-control study.

Were the cases clearly defined?
Were the cases representative of a defined population?
How were the controls selected and were they drawn from the same population as the cases?
Were study measures identical for cases and controls?
Were study measures objective or subjective and is recall bias likely if they were subjective?

Key methodological points to consider in the appraisal of a cross-sectional study.

Was the study sample clearly defined?
Was a representative sample achieved (e.g. was the response rate sufficiently high)?
Were all relevant exposures, potential confounding factors and outcomes measured accurately?
Were patients with a wide range of severity of disease assessed?

Key methodological points to consider in the appraisal of a study of diagnostic accuracy.

Does the sample of patients represent the full spectrum of patients with and without the diagnosis of interest?
Was there a comparison with an appropriate 'gold-standard' test?
Did all patients receive both the test under evaluation and the same 'gold-standard' test?
Were the tests performed independently with blinding of assessors to the results of the 'gold-standard' test?
Were the cut-offs that were used to classify patients as having a positive test result clearly described?

SPMIM Stats

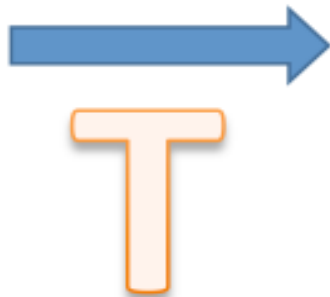
Evaluating Diagnosis

NOTE: Before reading further please read thoroughly the concepts of Odds & Probabilities explained in Advanced Statistics module.

Drawing a 2X2 table: A vital step in solving critical appraisal exercises related to the exam is mastering the technique of drawing a 2X2 table using information from journal articles. A standard method is followed throughout SPMM Stats notes. Please note that a wrongly drawn table will produce incorrect results. Studies that evaluate a diagnostic test (often a screening instrument in psychiatry e.g. CAGE etc) generally evaluate the relationship between two phenomena – being positive for the standard diagnostic test vs. being positive for the evaluated screening test. These are generally cross sectional studies where the whole study sample can be classified into 4 groups A, B, C and D as follows

- True positive = A, true negative = D
- False positive = B, false negative = C

	Diseased (Gold std positive)	No disease (gold std negative)	
Test positive	A	B	A+B
Test negative	C	D	C+D
Total	A+C	B+D	A+B+C+D



Total number of subjects in the study is (A+B+C+D). This can be classified using the results of a gold standard test (comparator) and results of the test being evaluated (often a screening test). Some subjects will show a positive result for the gold standard test (the ones who are 'really' diseased: A+C) and others will not have the disease and so will be negative for the gold standard test (B+D). Among those who are really diseased i.e. the A+C group, only some will show positive result for the screening test. They are called true positives (A). Those who show negative result for the screening test but are really diseased constitute the false negative group (C). Among those who are not diseased i.e. the B+D group, only some will show a negative result for the screening test. They are called true negatives (D). Those who do not show positive result for the screening test but are really diseased constitute the false positive group (B).

Letter T below the 2X2 table reminds us to use **T**est measures in the first column.

Two groups - Two properties:

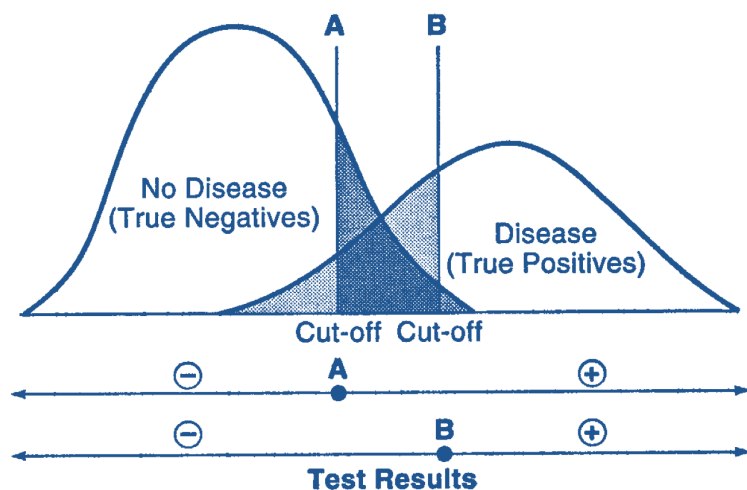
The definitive groups in the sample are those who are really diseased and those who are not. Let us define the properties of these definitive groups. In general, **sensitivity** gives the usefulness of a test in the context of truly diseased population; **specificity** gives the usefulness of a test in the context of non-diseased (or false presentations) population

- **Sensitivity** = true positive/ total diseased = $A / A+C$
 - Among a group of diseased individuals ($A+C$), how many can be correctly picked up by the test (A)?
- **Specificity** = true negative / total non diseased = $D / B+D$
 - Among a group of healthy individuals ($B+D$), how many can be correctly excluded by the test (D)?
- An allied measure is called as **Accuracy** = all correct 'hits' / total hits = true positive + true negative / total population = $A + D / A+B+C+D$

The maximum values of all the three measures are 100% (= 1).

The concept of cut-off:

Most screening tests measure some continuous biological parameter, whose level determines test positivity or negativity. This level is called a cut-off. By varying the cut-off levels, noticeable changes can be seen in the sensitivity and specificity as shown in the following figure.



Highly sensitive test	Highly specific test
Will pick up the cases that show even slightest of the evidence	Will pick up cases only if definitive evidence is noted
Makes more false positive inclusions	Makes more false negative exclusions
Preferred if a diagnosis should not be missed, but 'over-diagnosis' is not that harmful	Preferred if missing some cases is not bad, but wrong labeling is bad/costly
S N Out: highly sensitive test if NEGATIVE helps to rule out a disease	S P In: highly specific test if POSITIVE helps to rule in a disease

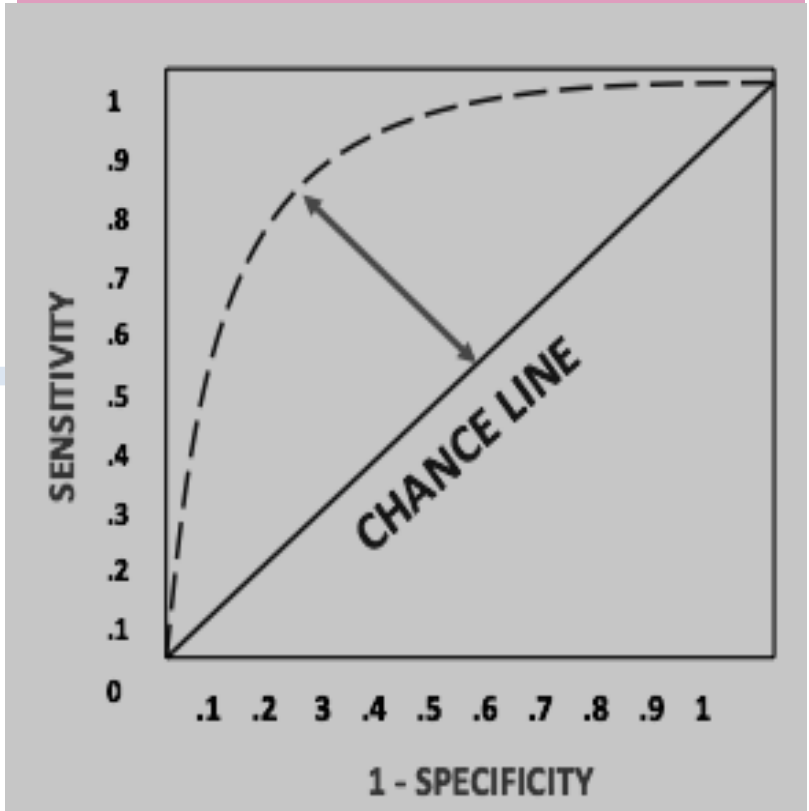
Consider the above diagram as a plot of number of individuals against prolactin level in serum, the first curve refers to all those who have low levels which are normal and the curve on right side refers to those with Hyperprolactinaemia. If the cut-off prolactin level is chosen at A then more persons will be diagnosed (high false positive, low specificity) but most of those who are truly hyperprolactinaemic will be diagnosed too (high sensitivity). If cut-off level is chosen at B then fewer persons will be diagnosed (low true positive, low sensitivity) but most of those who are 'normal' will be excluded (low false positive, high specificity) at the expense of some who are actually hyperprolactinemic (high false negative).

Receiver operator curve is useful in

1. Choosing between two diagnostic tests of different sensitivity and specificity rates
2. Choosing a cut off point for making a diagnosis.

- ROC analysis is used to select the optimal cut point to dichotomize a continuous scale.
- The usual choice of cut points minimizes the overall number of false positive and false negative errors.
- The cut point may be shifted if the cost of false positives is higher than that of false negatives, or vice versa.
- The accuracy of ROC analysis depends on the quality of the gold standard, which is usually far from perfect in psychiatry.
- Changing the purpose of the test (for example, from diagnosis to screening) requires a shift in cut points.
- A cut off point that is ideal for one group may be less than ideal for another.
- It shows the trade-off between sensitivity and specificity (any increase in sensitivity will be accompanied by a decrease in specificity).
- The closer the curve follows the left-hand border and then the top border of the ROC space, the more accurate the test.
- The closer the curve comes to the 45-degree diagonal (the chance line) of the ROC space, the less accurate the test.
- The area under the curve is a measure of test **accuracy**. (*Canadian Journal of Psychiatry*, 52, 2, 2007)

→ **ROC curve:** By plotting the true positive rates (sensitivity) on y axis and corresponding false positive rates (1-specificity) on the x axis the curve is obtained. The knee or elbow of the curve is a cut off point with balance between sensitivity and specificity. Among 2 different test curves, the one closer to left upper corner is the better screening test. (Milgrom, J et al (2005) *Screening for postnatal depression in routine primary care: properties of the Edinburgh Postnatal Depression Scale in an Australian sample*. *Aus NZ Journal of Psychiatry* 39 (9) ,



Likelihood ratios:

- Sensitivity and specificity have limited use in clinical practice. They are characteristics of a test itself; not of the clinical population tested. Hence on their own, they do not inform a clinician about the patient in question. A more useful approach is to combine the sensitivity and specificity results into single measures that tell us how much more likely a positive or negative test result is to have come from someone with a disease than from someone without it. These are known as the likelihood ratios for a positive test (LR+) and for a negative test (LR-). It is the value a test has in predicting the presence of a disorder.
- **LR+** = likelihood of testing positive rightly/wrongly - $(A/A+C)/(B/B+D)$.
- **LR-** = Likelihood of testing negative rightly/wrongly - $(C/A+C)/(D/B+D)$
- Expressed otherwise (NOTE: Always sensitivity on top) –
 - $LR+ = \text{sens} / 1 - \text{spec}$
 - $LR- = 1 - \text{sens} / \text{spec}$
- LR+ is usually >1 for most clinically available tests.
- **LR >10 or <0.1** can be used conclusively to use or avoid a diagnostic tool respectively.
- Likelihood of a diagnosis depends on the prevalence or prior probability of the disease before applying a test.
- **Pretest probability** = prevalence in the studied population = $(A+C)/(A+B+C+D)$
- From this post-test probability can be derived. One cannot multiply the *probability* by a likelihood *ratio*; probabilities need to be converted to odds before the likelihood ratio can be used. (Refer to 'Odds and Probabilities' chapter in Advanced Statistics module)

Likelihood ratios

- Tests often don't give clear +/- results (dimensional results more often noted)
- We often use multiple diagnostic tools
- How much more likely is a positive test to be found in a person with the condition than in a person without it?
 - LR of + test
- How much more likely is a negative test to be found in a person without the condition than in a person with it?
 - LR of - test

Method 1: Using pre-test odds to calculate post-test probability

1. Step 1: Obtain pretest probability. This is often the prevalence of a disease in the sample.
2. Step 2: Convert pretest probability to pretest odds using the formula

$$\text{Pretest odds} = \text{prob} / 1 - \text{prob}$$

$$= \text{diseased} / \text{not diseased}$$

$$= (A+C)/(B+D)$$

3. Step 3: Convert pretest odds to post test odds using

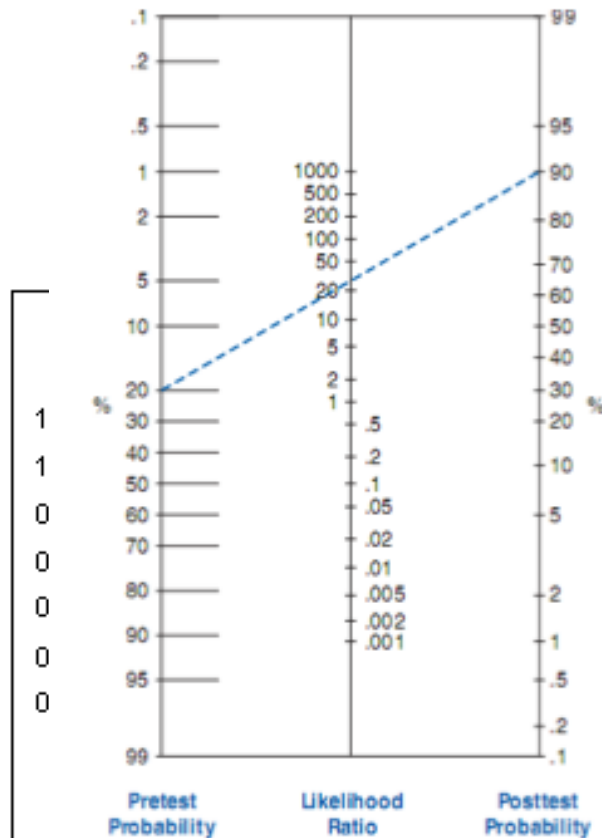
$$\text{Post test odds} = \text{LR} * \text{Pretest odds}$$

4. Step 4: Convert posttest odds to post test probability using

$$\text{Post test probability} = \text{posttest odds} / 1 + \text{posttest odds}$$

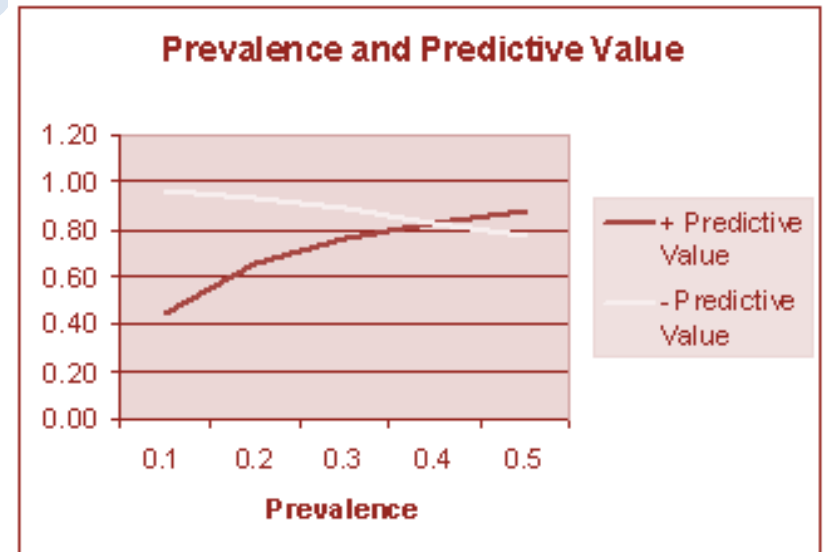
Method 2: Using Bayesian nomogram (Fagan) to calculate post-test probability

1. Step 1: Obtain pretest probability. This is often the prevalence of a disease in the sample.
2. Step 2: Obtain likelihood ratio (LR+ / LR-).
3. Step 3: Draw a line using straight edge across the two values available.
4. Step 4: Post test probability is the value obtained on the other side of the nomogram.



Predictive values:

- **Positive predictive value PPV** – true positive / total test positive – more useful measure to apply for individual patient's result – informs the chance of having the disease if tested positive = $A / A+B$. This increases with prevalence.
- **Negative predictive value NPV** – true negative / total test negative – informs the chance of not having the disease if tested negative = $D / C+D$. This decreases with increasing prevalence.



Evaluating Causation

NOTE: Before reading this further, please read the concepts of Odds & Probabilities explained in Advanced Statistics module.

Drawing a 2X2 table: A vital step in solving critical appraisal exercises related to the exam is mastering the technique of drawing a 2X2 table using information from journal articles. A standard method is followed throughout SPMM Stats notes. Please note that a wrongly drawn table will produce incorrect results.

Studies that evaluate causation (often a screening instrument in psychiatry e.g. CAGE, etc.) generally evaluate the relationship between two phenomena – exposure and outcome. These are generally either case-control designs or cohort designs.

Case-control studies:

The total number of subjects in the study is (A+B+C+D). In case-control studies, the cases and controls are the definitive groups – hence the total sample can be classified accordingly to cases (A+C) and Controls (B+D). Among A+C cases, some would have been exposed to the causative agent under study (A) while others will remain unexposed (C). Similarly, among B+D controls, some would have been exposed (B) while the others will remain unexposed (D).

	Cases	Controls	
Exposed	A	B	A+B
Non exposed	C	D	C+D
Total	A+C	B+D	A+B+C+D



Two groups: Four measures

The two definitive groups can be described according to the odds and probabilities of exposure.

Case exposure rate = $A/A+C$

Control exposure rate = $B/B+D$

Odds of exposure in cases: = number of exposed cases / number of non-exposed cases = A/C

Odds of exposure in controls: = number of exposed controls / number of non-exposed controls = B/D

Letter E below the 2X2 table reminds us to use **Exposure** measures in the first column.

Odds ratio

This is given by the ratio of the odds of being exposed if you are a case rather than a control.

OR = odds of exposure in cases/ odds of exposure in controls = $(A/C)/(B/D) = AD/BC$

Note that if a disease is very rare (prevalence < 5% arbitrarily), then odds ratio approximates RR.

OR can have values only between 0 and infinity (no negative values). Hence to calculate confidence intervals one needs to use log transformations.

Cross product

	Cases	Controls	
Exposed	A	B	A+B
Non exposed	C	D	C+D
Total	A+C	B+D	A+B+C+D

AD/BC is a simple cross product in a 2X2 table from a case-control study

	Disease develops	No disease	
Exposed cohort	A	B	A+B
Non exposed	C	D	C+D
Total	A+C	B+D	A+B+C+D



Cohort studies:

An important advantage of cohort designs is that we can calculate **risk rates** from these studies. The risk is the probability of an event happening – in other words, the incidence rate of an outcome when a population of interest is observed for a period of time. Incidence needs prospective observation – hence cannot be obtained from case-control designs.

The total number of subjects in the study is $(A+B+C+D)$. In cohort studies, the exposed, and the non-exposed are the definitive groups – hence the total sample can be classified accordingly to exposure status as: exposed $(A+B)$ and non-exposed $(C+D)$. Among $A+B$ exposed individuals, some will develop the expected disease (A) while others will remain healthy (B). Similarly, among $C+D$ non-exposed individuals, many will remain healthy (D) while some may become unwell (C).

Letter E below the 2X2 table reminds us to use **Exposure** measures in the first column.

Two groups: Four measures

The two definitive groups can be described according to the odds and probabilities of developing the expected disease.

- Absolute risk of disease in the exposed = $A/A+B$
- Absolute risk of disease in the non-exposed = $C/C+D$
- Odds of finding the disease among the exposed: = number of diseased individuals / number of non-diseased among the exposed = A/B
- Odds of finding the disease among the non-exposed: = number of diseased individuals / number of non-diseased among the non-exposed = C/D

OR = odds of the disease among the exposed/ odds of the disease among the non-exposed = $(A/B)/(C/D) = AD/BC$. Note that this measure is the same as the OR derived from a case-control design.

Risk rates:

Risk difference: This is the difference between two absolute risks described above in the two groups

$$RD = AR_{\text{exposed}} - AR_{\text{non-exposed}}$$

This is also called as absolute risk reduction (ARR: preferred term in clinical trials) or the **Attributable Risk** (preferred term in causal studies).

Relative risk: The most important final estimate from most cohort studies is the relative risk. In cohort studies, **RR** can be calculated as the ratio of the disease (outcome) in the exposed to the disease in non-exposed

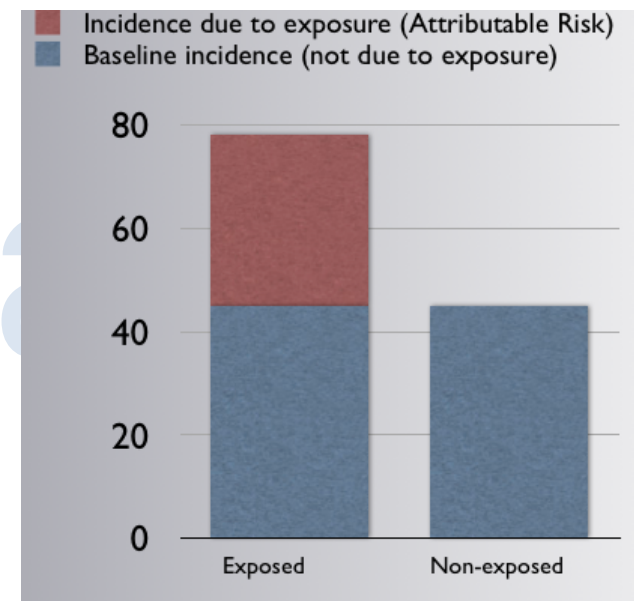
$$RR = [A / (A+B)] / [C / (C+D)]$$

RR is the ratio between the probability of developing the disease if you are exposed to the probability of developing the disease if you are not. It is clinically more meaningful measure than the OR. It is also called as the **risk ratio**.

Applying to local population:

Population Attributable Risk: This is also called as the *risk daddy* by epidemiologists. It is a very useful public health measure. PAR is obtained by multiplying Attributable Risk by the proportion of the total population exposed to the studied causative agent. PAR provides the number of cases that could be eliminated by eradicating the exposure.

$$PAR = ARR \times \text{Proportion exposed}$$



OR and RR

- Consider $RR = \frac{A/(A+B)}{C/(C+D)}$
- If very few of the exposed develop the disease then the fraction $A/A+B$ will be more or less = A/B ; similarly if very few of the non exposed develop the disease then the fraction $C/C+D$ will be more or less = C/D .
- Now look at the equation

$$\frac{\left(\frac{a}{a+b}\right)}{\left(\frac{c}{c+d}\right)} \approx \frac{\left(\frac{a}{b}\right)}{\left(\frac{c}{d}\right)} = AD/BC = \text{odds ratio}$$

OR and RR:

Under certain situations, odds ratios become equivalent numerically to relative risks. Such a circumstance occurs when the probability of an event occurring is low. This has been called the rare disease assumption, with a probability (or prevalence) of less than 5% being a commonly used benchmark. However, when an event occurs more commonly, the odds ratio increases dramatically, even as the relative risk remains constant as shown below. Apart from case-control studies (and cohort studies), odds ratios are routinely used in analyses of dichotomous outcomes in clinical trials as they greatly facilitate statistical analyses.

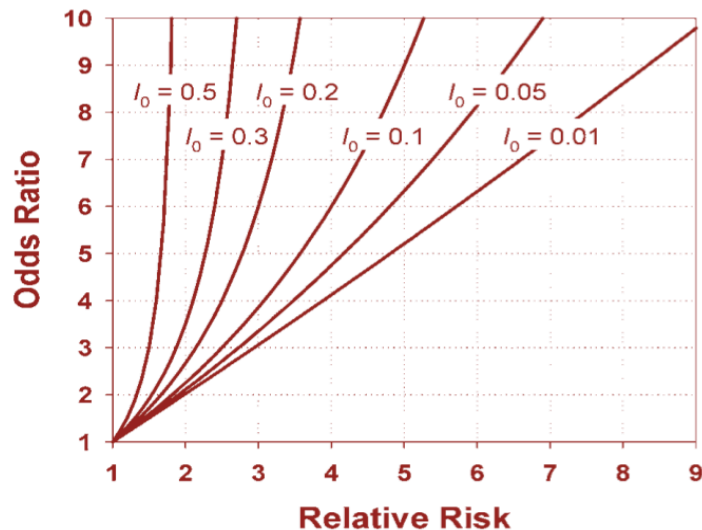


Figure 1 Relationship between Odds Ratio and Relative Risk for various incidence rates

Measures	Range of values
Probabilities (exposure rates)	0-1 (0 Indicates that there is no exposure in the group; 1 indicates that everyone has been exposed)
Absolute risks	0-1 (0 indicates that there is none with the disease in a group; 1 indicates that everyone in a group has the disease)
Attributable Risk	-1 to +1 (-1 indicates that the risk is greater in the non exposed group)
Relative risk	0 to infinity (<1 indicates that the risk in the exposed is less than the risk in the non-exposed; =1 means that the risks are equal in both groups, so there is no causal association)
Odds	0 to infinity
Odds ratio	0 to infinity

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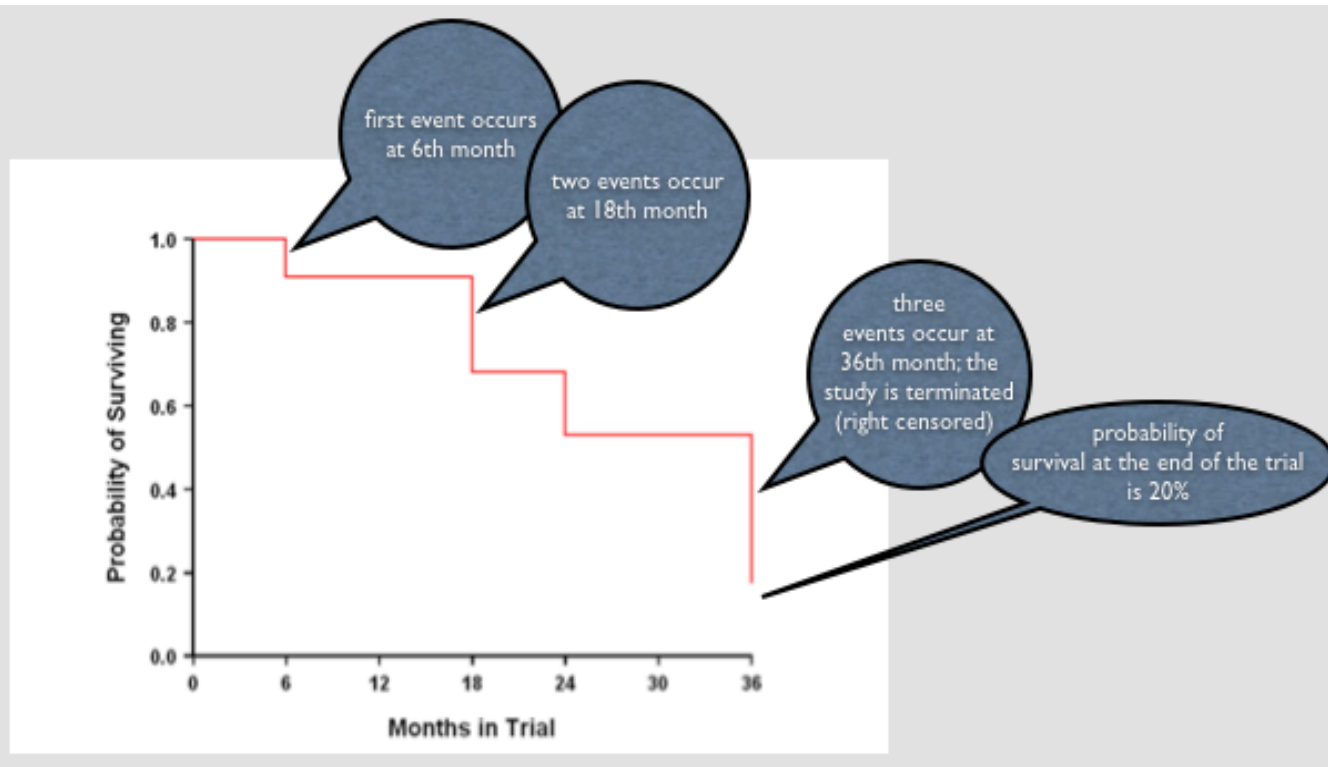
<http://jama.jamanetwork.com/article.aspx?articleid=188182>

<http://www.ncbi.nlm.nih.gov/pubmed/16785380>

Evaluating Prognosis

Both cohort studies and RCTs are used for estimating the prognosis of clinical samples. In chronic diseases such as schizophrenia, an important survival question relates to the risk of relapse/rehospitalisation when taking treatment. In addition, many events of interest such as medication discontinuation, self-harm and violence can be quantified when time-to-event is taken into consideration.

A simple graphical method for doing this is plotting the time along x-axis and proportion surviving (i.e. not attaining) 'an event' is plotted along the y-axis. The event could be death, treatment response, relapse, etc. The cumulative probability of an individual remaining free of the event is displayed at each point. These curves have a stepped appearance, with each step taking place at a time of observation. These are called **Kaplan-Meier Survival Curves**.



- ♣ If the study runs for a 5-year period, then not all subjects would have reached the outcome by that time. Hence, these observations are **censored**. They are called **left censored** if the observations began well after some events have occurred (missed past events) or **right-censored** if the observation was terminated before all events could be observed (missed future events).
- ♣ Due to censoring we often do not know the actual survival times beyond the end of the study period; so we cannot calculate the mean survival time of the whole group. However, we can determine median survival time, which corresponds to the survival time of 50 per cent of the observed sample. In the above figure, the median survival time is 24 months. At this time, half of the patients still survived. Obviously the median survival time can be calculated only if the observation was long enough for enough events to take place in half of the sample so that the Kaplan-Meier curve goes down far enough.
- ♣ Comparison of survival rates of two samples can be undertaken using **log-rank test**. It is a non-parametric test to analyze the difference between the survival curves of two groups. But the log-rank test cannot be used to account or adjust for effects of other variables such as confounders (because it is a univariate test). To study multiple variables one needs multivariate logistic **Cox regression analysis** (called as proportional hazard as it assumes that the effects of multiple variables are constant over a period of time – it is a semiparametric method). See the table below for further methods of analyzing survival data
- ♣ Log-rank test only informs whether significant differences are present or not by comparing two survival curves. It does not measure the magnitude of such differences. **Hazard ratio** gives the size of such differences.
- ♣ **Mantel-Haenszel test** could be used in life tables to study the effect of two binary independent variables (e.g. male vs. female and treatment vs. placebo) on the outcome.

Number of independent variables (categorical)	Nature of variable	Method
1	Continuous (e.g. days, years, etc)	Kaplan-Meier curve with Log-rank test
>1 (multivariate)	Continuous	Cox's proportional hazard
2	Continuous	Mantel-Haenszel test

Evaluating a prognostic study:

The following questions help appraising a prognostic study:

1. Are the results valid?

Was the sample of patients representative?

First, determine whether patients pass through some sort of filter before entering the study. If they do, the result is likely a sample that is systematically different from the underlying population of interest.

Were the patients sufficiently similar to each other with respect to prognostic factors?

Including different stages of disease in the cohort will make the survival estimates of the group not applicable to each group member.

Was follow-up sufficiently complete?

The larger the number of patients whose fate is unknown relative to the number who had the event of interest, the greater the threat to the validity.

Were outcome criteria objective and unbiased?

Outcome events may be objective and easily measured (eg, death), require some judgment (eg, myocardial infarction), or require considerable judgment and effort to measure (eg, disability, quality of life). Investigators should clearly specify and define their target outcomes and, whenever possible, they should base their criteria on objective measures.

2. What are the results?

How likely are the outcomes over time?

This is given by the Kaplan-Meier curve. Note that the median survival time may be misleading in certain instances as shown in the accompanying graph.

How precise are the estimates of likelihood?

In most survival curves, the earlier stages of follow-up usually include results from more patients than the later periods (owing to losses to follow-up and because patients are not enrolled in the study at the same time), which means that the survival curves are more precise in the earlier periods, indicated by narrower confidence bands around the left-hand parts of the curve

3. How can I apply the results to patient care?

Were the study patients and the clinical context similar to those in my practice?

Was the follow-up sufficiently long?

Depending on the nature of the outcome of interest, this must be addressed.

Why survival curves?

- Recruitment takes 2-3 years
- Patients followed till the end
- If minimum follow-up is 5 years, then significant loss – attrition
- Many enter late, many do not reach end point
- Focus is on time to end-point

ENDPOINTS

- Death during trial
- Loss to follow-up
- Still alive at the end (right censor)
- Adverse event
- Discontinuation of treatment (CATIE)

Key methodological points to consider in the appraisal of a cohort study

Is the study prospective or retrospective?

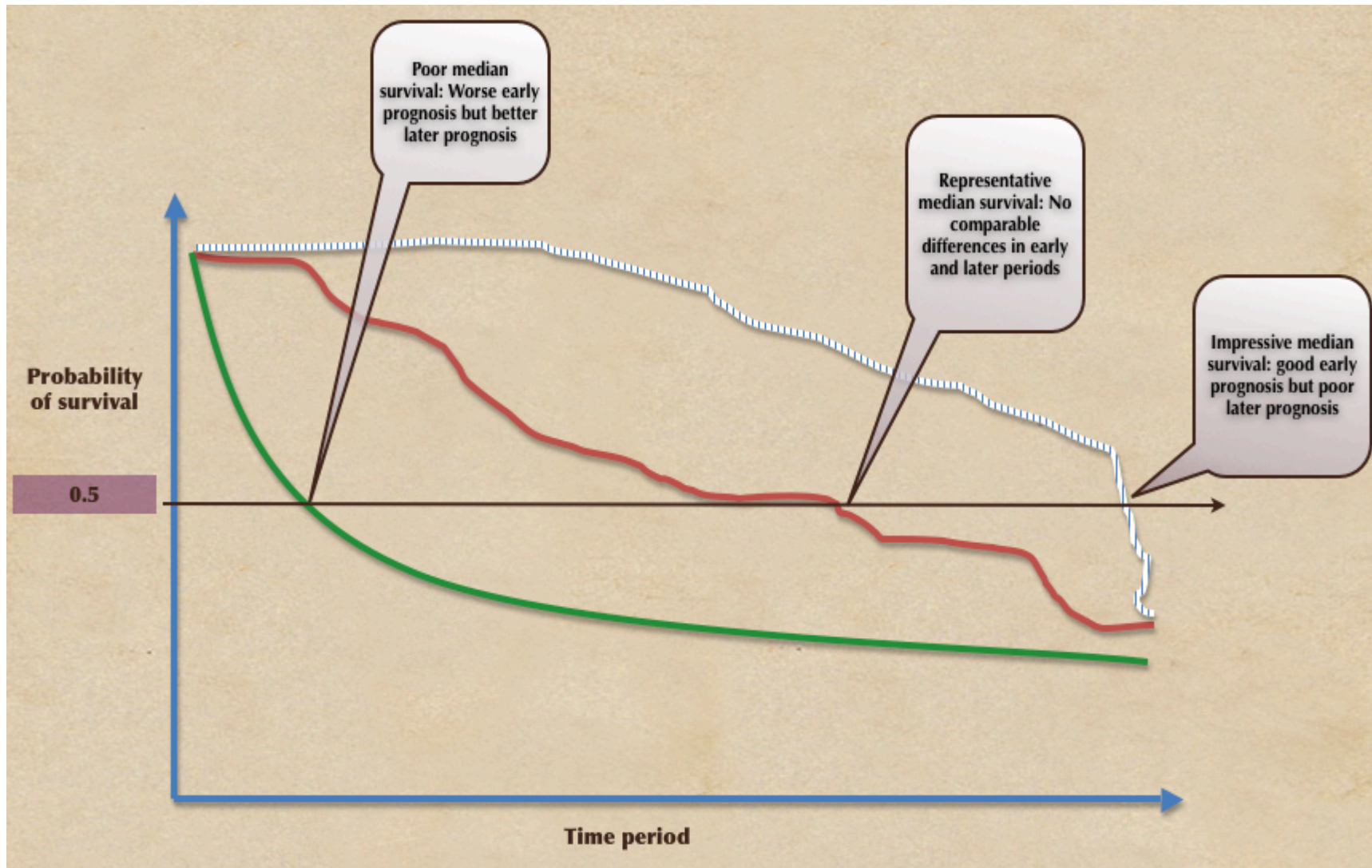
Is the cohort representative of a defined group or population?

Were all important confounding factors identified?

Were all important exposures and/or treatments, potential confounding factors and outcomes measured accurately and objectively in all members of the cohort?

Were there important losses to follow-up?

Were participants followed up for a sufficient length of time?



Three different curves with variable median survival times but comparable survival at the end period.

Evaluating Therapy

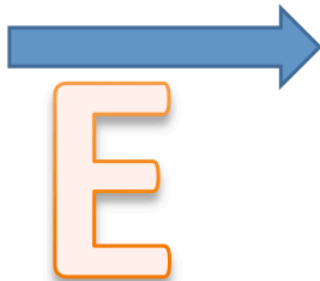
NOTE: In this chapter, evaluation of both beneficial effects (Therapy) and adverse effects (Harm) from a RCT is considered.

Before reading further please read thoroughly the concepts of Odds & Probabilities explained in Advanced Statistics module.

Drawing a 2X2 table: A vital step in solving critical appraisal exercises related to the exam is mastering the technique of drawing a 2X2 table using information from journal articles. A standard method is followed throughout SPMM Stats notes. Please note that a wrongly drawn table will produce incorrect results. Studies that evaluate therapy generally evaluate the relationship between two phenomena – exposure to an intervention and the resultant outcome. These are generally RCTs.

Total number of subjects in the study is $(A+B+C+D)$. In RCTs, nature of the intervention defines the two groups. Those assigned to the active intervention (treatment group) and placebo or standard intervention (placebo group) form the two definitive groups – hence the total sample can be classified accordingly to Treatment $(A+B)$ and Placebo $(C+D)$. Among $A+B$ subjects who receive treatment, some will experience the outcome of interest (benefit or harm) (A) , while others will not experience this event (B) . Similarly, among $C+D$ subjects taking placebo, some will experience the outcome of interest (C) while the others will remain 'uneventful' (D) .

	Outcome present	Outcome absent	
Treatment	A	B	A+B
Placebo	C	D	C+D
Total	A+C	B+D	A+B+C+D



Two groups: Four measures

The two definitive groups can be described according to the odds and probabilities of developing the expected outcome ('event').

- CER = Control event rate = Frequency of event (positive outcome in case of treatment trials; adverse event in case of harm evaluation) in the placebo control group = $C/(C+D)$
- EER = experimental event rate = Frequency of event (positive outcome in case of treatment trials; adverse event in case of harm evaluation) in the treatment group = $A/(A+B)$

Odds of
the event of interest in the Treatment group:

Letter E below the 2X2 table reminds us to use **Exposure to intervention** (treatment vs. placebo control) in the first column.

= Number of subjects developing the outcome / number of subjects without the outcome in the Treatment group = A/B

- Odds of the event of interest in the Placebo group: = number of subjects developing the outcome / number of subjects without the outcome in the Placebo group = C/D

Odds Ratio = odds of the outcome in the Treatment group/ odds of the outcome in the Placebo group = (A/B)/(C/D) = AD/BC. Note that this measure is the same as the cross product OR derived from a case control or cohort design.

Risk rates:

Absolute Benefit Increase: This is also called as Absolute Risk Reduction. This is the difference between two event rates described above in the two groups

$$\text{ABI (or ARR)} = \text{EER} - \text{CER}$$

Relative risk: = Ratio between risk of event (the outcome) in treatment group and risk of event in control group

$$\text{RR} = [A / (A+B)] / [C / (C+D)] = \text{EER} / \text{CER}$$

Relative Benefit Increase: This is also called as Relative Risk Reduction. It is the Absolute Benefit Increase expressed as a proportion of risk in the control group.

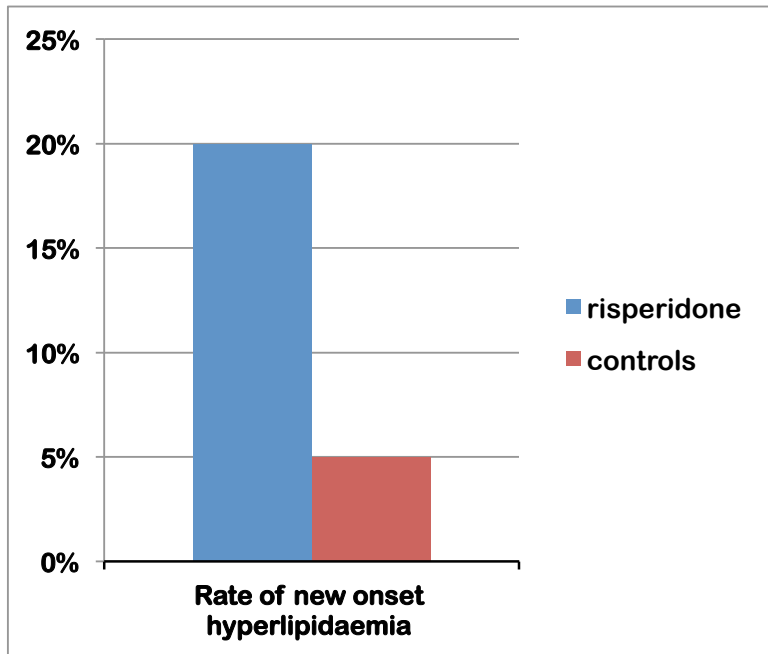
$$\text{RBI} = \text{ABI} / \text{CER} = (\text{EER} - \text{CER}) / \text{CER} = \text{RR} - 1$$

NNT & NNH:

Number needed to treat: the number of patients who would have to receive the treatment for one of them to benefit; calculated as 100 divided by the absolute risk reduction expressed as a percentage (or 1 divided by the absolute risk reduction expressed as a proportion)

Number needed to harm: the number of patients who would have to receive the treatment for one of them to experience an adverse effect; calculated as 100 divided by the absolute risk increase expressed as a percentage (or 1 divided by the absolute risk increase expressed as a proportion)

$$\text{NNT or NNH} = \text{Number of patients that need to be treated for one additional beneficial or adverse outcome} = 1 / \text{ABI}$$



Consider the graph here: There are two groups described - A group taking Risperidone (experimental group) and the group taking placebo (placebo control group). The outcome of interest is new onset hyperlipidaemia.

Control event rate is 5%.

Experimental event rate is 20%.

Absolute Risk Reduction (ARR or ABI) is $20\% - 5\% = 15\%$ (NOTE: do not drop the percentage)

Number needed to harm: $100/15 = 7$ (NOTE: rounded up to nearest higher integer)

Relative risk is $EER/CER = \frac{20\%}{5\%} = 4$

Relative risk reduction is $1 - 4 = 3$ (NOTE: drop the negative sign); or $(EER - CER) / CER = \frac{15\%}{5\%} = 3$

As hypercholesterolemia is an adverse event, we calculate NNH here. If the outcome of interest were a beneficial treatment effect, we would prefer using the term NNT. $NNT < 10$ is taken as significant value arbitrarily – the clinical significance may vary according to the nature of the intervention. The perfect NNT is 1, where everyone gets better with treatment and no one gets better with control. NNTs of close to 1 are seen with, for instance, antibiotic treatment of susceptible organisms. NNTs of 2-5 represent effective therapies, but NNTs of 20 or more may be useful only in prophylaxis. (Bandolier, Feb 2003)

Applying to local population:

A major difference between trial participants and your clinical population is the differences in baseline risk rates. You may be working in an affluent area where the baseline rate for the outcome of interest is much higher or lower – e.g. due to a healthy lifestyle, your patient sample may have a very low risk of hypercholesterolemia at baseline. So, when adapting the results of a RCT to your practice, you need to change the

Control Event Rate (baseline risk in the group without the intervention). The new CER for your clinical sample is called PEER: patient expected event rate.

- PEER = patient expected event rate is control event rate in local population in which the intervention is planned. Using this PEER, we can obtain locally useful ABI called ABI(new).

$$\text{ABI}(\text{new}) = \text{RBI} \times \text{PEER}$$

- We can also get new NNT

$$\text{NNT}(\text{New}) = 1/\text{ABI}(\text{new})$$

- Another way of calculating NNT (new) is via F fraction, which is the susceptibility of a local population to a particular outcome. For example, if you think half of your clinical population gets hypercholesterolemia even without using risperidone, then F ratio is 0.5

$$\text{NNT}(\text{new}) = \text{NNT}(\text{old}) / \text{F value}$$

(You will be given the PEER value or the F value to make calculations in the exam)

Worked example:

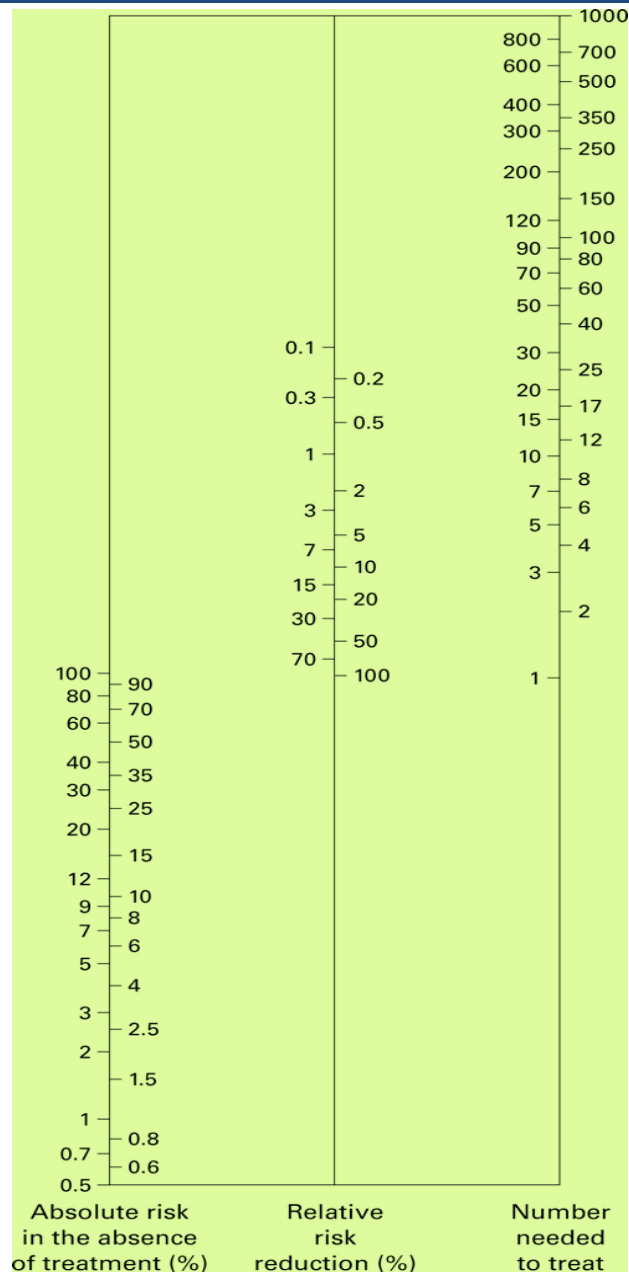
The following data is from a RCT comparing CBT and IPT (*The British Journal of Psychiatry* (2007) 190: 496-502).

The secondary outcome measure was derived as a categorical outcome of a 60% or greater improvement in MADRS score, and any statistical difference between number of responders in two groups were analyzed. 1. Calculate the probability of getting better on CBT compared to IPT; 2. The odds of getting better on CBT alone. 3. How many patients of yours will need treatment for one additional patient to improve if the treatment response for depression in your specialist service is only half of what is given here?

Table 2 Primary and secondary outcomes by therapy in the intention-to-treat sample

	IPT (n=91)	CBT (n=86)	
Participants completing therapy, n (%)	83 (91)	76 (88)	$\chi^2=0.39$, NS
Participants commencing maintenance phase, n (%)	74 (81)	71 (83)	$\chi^2=0.05$, NS
MADRS score: mean (s.d.)			
Baseline	23.3 (6.5)	24.4 (6.2)	$t=1.13$, NS
Final	12.6 (9.6)	10.5 (9.4)	$t=1.44$, NS
Primary outcome			
Improvement in MADRS score, % (s.d.)	49.5 (32.9)	58.9 (33.0)	$t=1.90$, 0.059
Secondary outcome			
Responders, n (%)	41 (45)	51 (59)	$\chi^2=3.60$, $P=0.058$

CBT, cognitive-behavioural therapy; IPT, interpersonal psychotherapy; MADRS, Montgomery-Åsberg Depression Rating Scale; NS, not significant.



NNT Nomogram

It is often useful to construct a second 2X2 table.

	Control (IPT)	CBT	Total
Responders	41	51	92
Non responders	50	35	85
	91	86	177

1. Probability of getting better on CBT compared to IPT is nothing but relative risk $RR = EER / CER$

$$EER = 51/86 = 0.59$$

$$CER = 41/91 = 0.45$$

$$\text{Hence } RR = 0.59/0.45 = 1.31$$

2. The odds of getting better on CBT = frequency of responders/ frequency of non responders undertaking CBT = $51/35 = 1.45$

3. To answer this question we need to know NNT (Old).

$$ABI \text{ (old)} = EER - CER = 0.59 - 0.45 = 0.14$$

$$NNT \text{ (old)} = 1 / ABI \text{ (old)} = 100/14 = 7.$$

$$NNT \text{ (new)} = NNT \text{ (old)} / F. \text{ Here } F = 0.5 \text{ (as outcome is only half likely)} = NNT \text{ (new)} = 14.$$

NNT can also be calculated from tables and nomogram.

Key methodological points to consider in the appraisal of a RCT

- Was the process of treatment allocation truly random?
- Would participants have been able to know or guess their treatment allocation?
- Were participants and researchers 'blinded' to participants' treatment group?
- Were outcomes assessed objectively?
- Were all participants who were randomly allocated a treatment accounted for in the final analysis?

Evaluating Meta-analysis

NOTE: Most meta-analyses evaluate Therapy or Harm. SO for the MRCPsych exam, calculations based on meta-analyses are carried out in the same fashion as calculations for RCTs. You can draw a 2X2 table from the information given in a forest plot if required. A worked out example is given below. Please read 'Evaluating Therapy' before reading this chapter.

Advantages of Meta-Analysis:

Well-conducted meta-analyses have a number of important advantages over individual RCTs. 1. Increased power: The primary advantage of meta-analyses is an increase in statistical power over that of individual RCTs. 2. Clarify the direction of literature: Often, in a given area of interest, various RCTs may provide contradictory results, Meta-analysis can either reveal an underlying unifying conclusion among seemingly contradictory study findings, or explore the reasons for contradictions.

Critically important aspects:

1. Methodologically rigorous meta-analytical techniques can be applied to suboptimal data leading to inconclusive results. No amount of statistical technique can improve the fundamental quality of the data being combined for the meta-analysis. This is sometimes referred to as garbage in-garbage out (GIGO) principle.
 2. Publication bias occurs as studies that produce a statistically significant result are much more likely to be published than "negative" trials. This can lead to overestimating the effect of the intervention in the meta-analysis. To avoid or reduce the impact of this bias, one must systematically search for negative trials for inclusion in the meta-analysis.
 3. Apples and Oranges problem: Clear inclusion criteria, the graphical presentation of the results in forest plots and heterogeneity statistics are important to avoid mixing widely different studies to give a final estimate. One cannot simply combine a cohort study result with a rigorous RCT result to give a final pooled estimate.
 4. Missing data. If the necessary parameters are not given in a study, it cannot be included in the meta-analysis. Actually one would like to cover all relevant studies, but although the authors of the original studies are contacted for missing information, not all of them can send this information.
 5. Differing statistic models. There are debates as to which statistical model is most suitable in which situation, even though often no relevant differences result from the choice in the case of robust data (apart from the different interpretation of absolute and relative risk reductions as explained above). It is a common practice to both fixed and random effect analyses in a given data set.
 6. Inappropriate subgroup analyses: Exploring subgroup findings is a common feature of meta-analyses, sometimes as a way of explaining a failure to find any overall effect. A subset of studies or subgroups of patients may be analysed separately. While this approach may offer insights that can be tested in further RCTs, caution should be exercised in interpretation. Despite the underlying studies being randomised, this randomization and the subsequent likely balance between treated and control groups does not extend to subgroups defined after the fact. Thus, there is great potential for confounding and misleading findings.
 7. Sensitivity analyses: (See 'Secondary Research' notes)
- The process of selection, inclusion and aggregation of data may affect the main findings in a meta-analysis. Hence meta-analysts commonly carry out some sensitivity analysis. This explores the ways in which the main findings are changed by varying the approach to aggregation. A good sensitivity analysis

should model the effect of excluding various categories of studies, e.g., unpublished studies or those of poor quality. It should also examine how consistent the results are across various subgroups. In meta-analyses without sensitivity analyses the reader has to make guesses about the likely impact of these important factors on the final results

Questions for appraisal of a meta-analysis:

(from Guyatt et al. JAMA series)

Is the meta-analysis based on a written protocol that clearly outlines research question, primary and secondary outcomes, and inclusion and exclusion criteria?

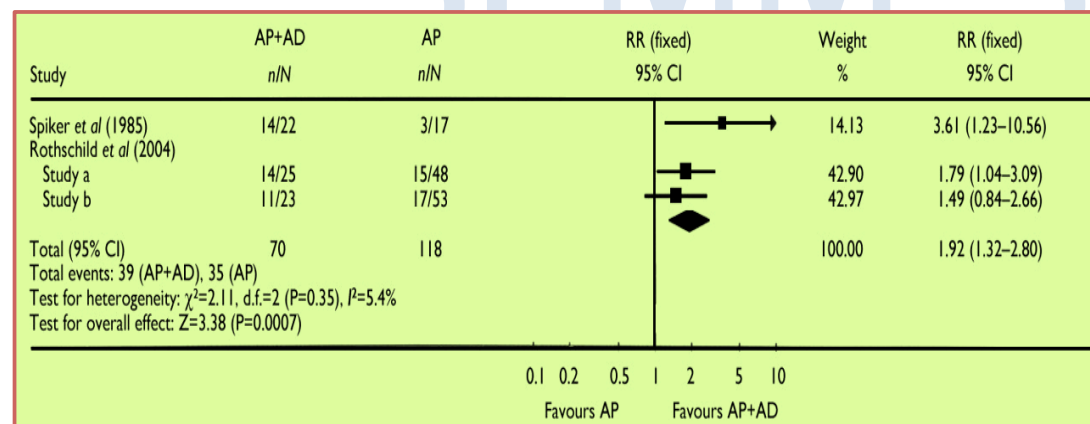
Has a thorough literature search been performed? Have different search engines (PubMed, Embase, Cochrane library, etc.) been used to identify relevant literature?

Did the authors look for unpublished data, for negative studies, and for publications in non-English languages to minimise retrieval, language, and publication bias?

Was a strategy to exclude individual studies clearly outlined in the publication?

Did two investigators independently perform the quality assessment of the individual studies?

Were sensitivity analyses performed?



Interpretation of Forest Plots

Efficacy of the combination of an antipsychotic (AP) plus an antidepressant (AD) (AP+AD – intervention evaluated) v. antipsychotic monotherapy (control) from WIJKSTRA, J. et al. Br J Psychiatry 2006;188:410-415

The above forest plot shows three studies – Spiker, Rothschild study a and Rothschild study b. Each study had two arms, AP only vs. AP+AD. The small n denotes number of people showing a pre-defined response to the treatment. The capital N denotes the number of participants in each arm. The point estimate plotted here is a Relative risk (RR).

As this is a ratio measure (similar to OR), the line of no difference is placed on the value 1. (Note that if the final estimate is standardised mean or Cohen's d effect size, then the line of no difference will be placed on 0). On either side of this line, we have the two possible interpretations of the study results. If a study's final point estimate (block or square) is on the right side of the line, then this will favour AP+AD. If it is on the left side, this will favour AP. As you can

see, all three studies have final estimated values on the right side. The horizontal line across each square denotes the confidence interval for the final estimate. If a study gives precise results (which happens especially if the sample size is large), then the length of this line will be shorter. Spiker study has a horizontal line whose right end cannot be shown in the plot – hence an arrow mark is placed: this is a sign of imprecision. Depending on the weight given to a study, the square will vary in size, bigger the square, greater the weight given to this study's results. The actual weight is also denoted as the 5th column. When you add up individual study weights, they will make to 100%. As study b crosses the line of no effect, its results are not significant statistically (though this study is heavily weighted: Weight does not depend on the magnitude of the point estimate). The diamond lozenge is the pooled result. It will have no horizontal line (in fact, a very small line that is gobbled up by the arms of the lozenge!) as the pooled results have highest precision. If this does not cross the line of no difference, we can be confident that the result of meta-analysis is decisive to some extent (at least statistically).

Recollect that when evaluating an RCT, we derive some important measures such as CER, EER and RR. (Refer to SPMM Stats 'Evaluating Therapy' notes). We can do the same from a meta-analysis too. For Spiker study, as AP is the control arm against which we do the comparison (similar to a placebo arm in most RCTs), n/N gives the Control Event Rate. The n/N for AP+AD gives the EER. As we know CER and EER, we can determine RR, RRR and ARR. From ARR, we can determine NNT. You can do the same for the remaining two Rothschild studies too.

The CER for the pooled analysis can be obtained by adding individual n and N. In the above plot, $(3 + 15 + 17 = 35)$ gives the pooled n for AP. $(17 + 48 + 53 = 118)$ gives the pooled N for AP. The pooled CER is 35/118. Similarly, the pooled EER is 39/70. (But pooled RR calculated in this manner can be different from the RR shown against the lozenge if the individual n/Ns are 'weighted' before getting added up).

SPMM Stats

Recommendations & Guidelines

While tackling a clinical problem, one may be guided by protocols or guidelines. **Protocols** carry a sense of strict adherence being required; **guidelines** are recommendations based on evidence.

Treatment recommendations can be of two types

1. Decision analysis trees
2. Clinical guidelines

Steps to be followed in producing a treatment recommendation:

1. Defining the decision (e.g. decision to switch antipsychotics when first episode of psychosis is not controlled even after 12 weeks of treatment)
2. Specify the possible outcomes (control of symptoms, no effect of a switch, an increase in side effects, noncompliance, occurrence of depression, etc.).
3. Using available evidence to specify the link between listed options and outcomes. (Olanzapine causes weight gain, lesser dropout rate for olanzapine and depots etc...)
4. Incorporate values/value judgements to decide on optimal course of action (female patient, poor lifestyle – weight gain is an important problem, etc.)
5. Consider local circumstances including resource availability and modify action if needed (no depot clinics locally, no CPNs, etc.).

Decision analysis refers to the provision of formal structure to the process of decision making. This is

represented as a flowchart or tree where on the left side decision to be analysed is placed; options are branched out in the centre, and clinical outcomes are placed on the right.

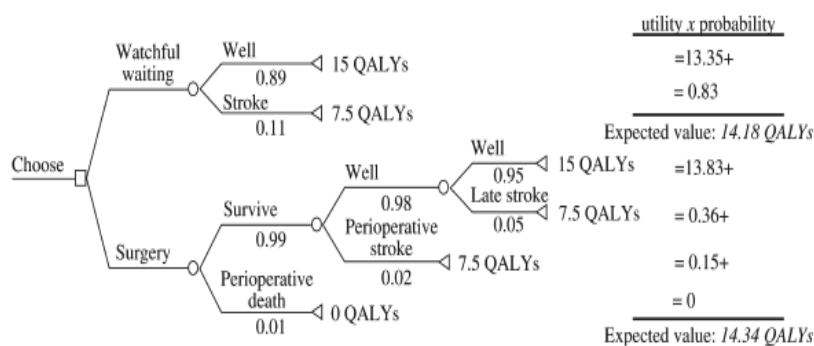


Fig 1. Calculating the expected values of watchful waiting and surgery for asymptomatic carotid stenosis (folding back the decision tree).

Decision chart: The squares on the left are called **decision nodes** while the circles on the right are called **chance nodes**. Once the analyst has drawn the tree he/she must generate quantitative estimates of the likelihood of events or probabilities – ranging from 0 –impossible to 1 – absolutely certain. Then the analyst places a utility value on each of the final outcome from 0 – death to 1 – full health (can be calculated using QALYs etc). In the final step the total value of each decision is calculated by cross product of utility and probability with summing of individual outcomes. Comparing the final values one can decide which the best course of action is. **Surgery 2003; 133:1-4**

Practice guidelines:

- ⊙ These are usually generated by a panel of stakeholders (e.g. NICE).
- ⊙ Guidelines based on same evidence base may vary in recommendations. This is because apart from being evidence based, guidance is also often value based, and local availability (cost) based.
- ⊙ One very important step in generating guidelines is having a systematically reviewed base of evidence on which recommendations can be built.

- ⊙ An explicit statement of values must be present in a proper guideline. (See NICE guideline summaries – these values and preferences get highlighted in background of most guidelines)
- ⊙ Note that it is not imperative that a National guideline will suit every local practice; hence the need for local adaptations.
- ⊙ In order to check if guidelines are valid we can ask 4 questions;
 1. Did all relevant patient groups, management options and outcomes get considered?
 2. Is there a proper systematic review?
 3. Is there an appropriate specification of values and preferences associated with outcomes?
 4. Do the authors indicate the strength of recommendations?

The **Hierarchy of evidence** based on the Oxford Centre for Evidence-based Medicine Levels of Evidence (May 2001) is as follows:

- I Evidence obtained from a systematic review of all relevant randomised control trials.
- II Evidence obtained from at least one properly designed randomised control trial.
- III-1 Evidence obtained from well-designed pseudo-randomised controlled trials (alternate allocation or some other method).
- III-2 Evidence obtained from comparative studies (including systematic reviews of such studies) with concurrent controls and allocation not randomised, cohort studies, case-control studies, or interrupted time series with a control group.
- III-3 Evidence obtained from comparative studies with historical control, two or more single-arm studies, or interrupted time series without a parallel control group.
- IV Evidence obtained from case-series, either post-test or pre-test and post-test.
- V Expert opinion without a critical appraisal, or based on physiology, bench research, or historically based clinical principles.

Usefulness of guidelines could be determined by asking three questions:

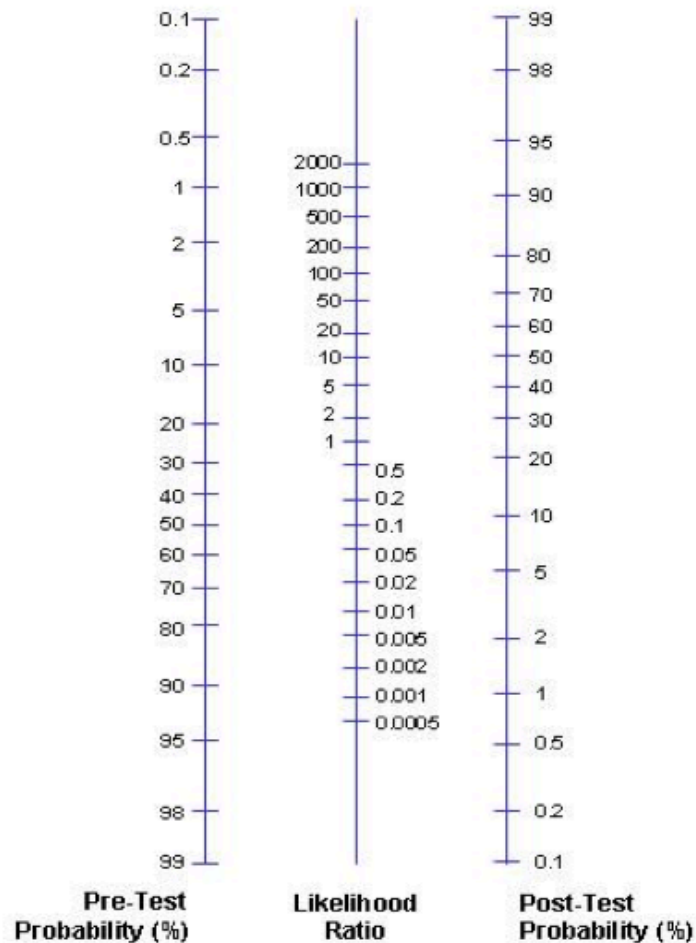
1. Who developed them?
 2. How were they disseminated?
 3. How were they implemented?
- The answers are; (Modified from Grimshaw & Russell 1993)

Effectiveness of guidelines	How were they developed?	How were they disseminated?	How were they implemented?
Highly effective	Developed by those who will use them	Specific educational intervention e.g. focused meeting, learning package with credits	Patient-specific reminder during consultation e.g. filed in the notes, etc.
Above average effectiveness	Intermediate – modified national or external guidelines	Continuing medical education e.g. lecture, lunch time meeting	Patient-specific feedback
Below average	External, not by those who use them but still local	Mailing target groups	General feedback e.g. gross audit
Very low effectiveness	External, national guidelines	Publishing in a journal.	General reminder

Guyatt, G & Rennie, D (ed). User's guides to the medical literature; a manual for evidence-based clinical practice. Page 175-196. AMA Press, 2002.

Grimshaw & Russell (1993) Effect of clinical guidelines on the medical practice. The Lancet 342, 1317-1322

Likelihood Ratio Nomogram



- The post-test probability is the **probability** that a patient, scoring positive on the test, actually has the disorder.
- Once we know the pre-test probability (prevalence) and the likelihood ratio, we should be able to find the post test probability from the chart.
- A straight line drawn through the pre-test probability and the likelihood ratio should yield the post test probability

Altman Nomogram

The total **sample size** required to detect an anticipated group difference (**effect**) with a specified ability (**power**) is obtained by joining the value of power on the right-hand axis and the standardized difference (size of effect) on the left-hand axis, using a straight line.

The intersection of this line with the upper part of the central platform gives the sample size required for the study to detect the difference with a preset probability threshold for false positives P of 0.05. The intersection with the lower part gives the sample size for P = 0.01.

The effect size is given by **standardized difference**. This is

$$\text{Standardised difference} = \frac{\text{Target difference}}{\text{Standard deviation}}$$

nothing but the target difference between the two groups studied, expressed in terms of the standard deviation of the variable across the study population.

The standard deviation is unlikely to be known in advance before beginning the study. But it may be possible to estimate it from searching literature or by doing a small pilot study.

Before starting a study, one may want to calculate required sample size by keeping constant p-value, power and estimated an expected effect size. Alternatively, in a published paper, one may wish to test the power of the study using the given sample size, found effect size and set p-value. Either way, one can make use of Altman's nomogram.

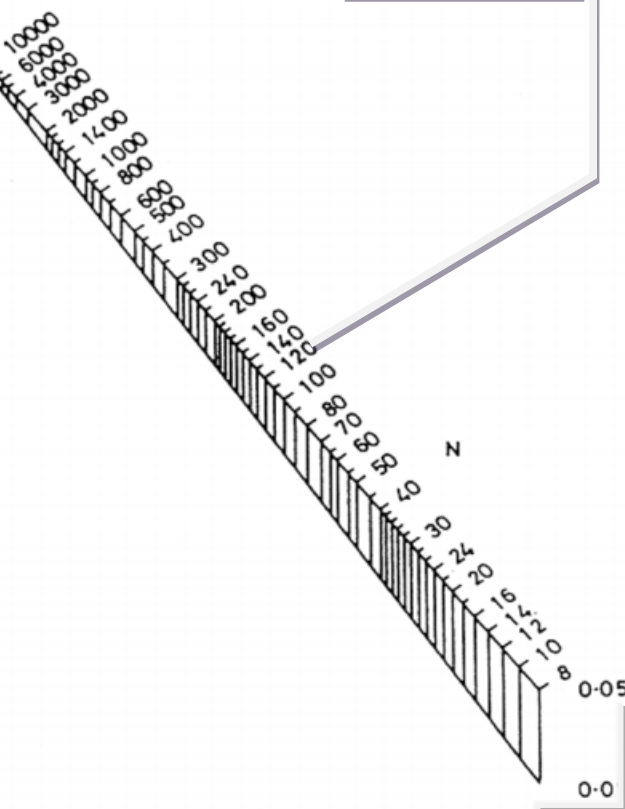
SPMM Memory Corner

Sample size depends on p value, power, expected difference and the expected variability of the outcome variable in the study population (Std. deviation).

STANDARDISED MEAN
DIFFERENCE (EFFECT SIZE)

0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2

REQUIRED
SAMPLE SIZE



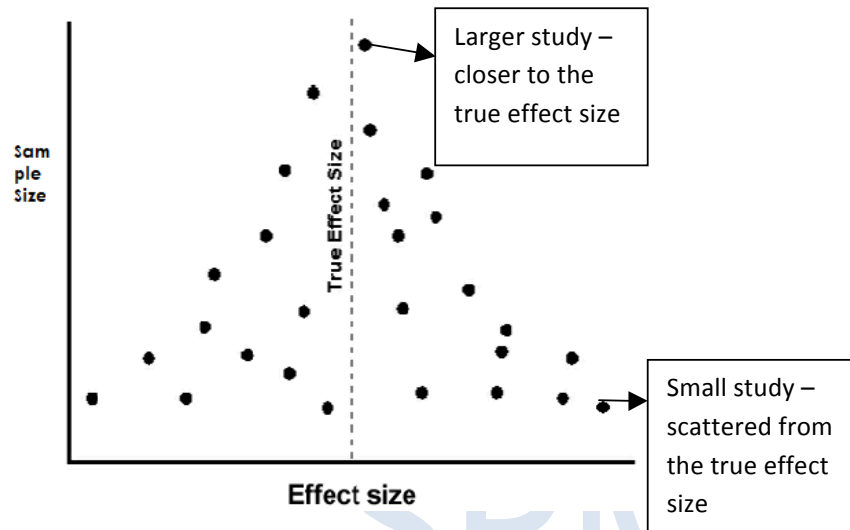
SIGNIFICANCE LEVELS
(P value)

SIG
LEVEL

POWER OF THE STUDY

0.995
0.99
0.98
0.97
0.96
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05

Funnel plot



A **funnel plot** is used to detect the presence of publication bias.

It is usually used in the context of a **meta-analysis**

A measure of **precision** (sample size or more commonly the Standard Error) is plotted along the y axis

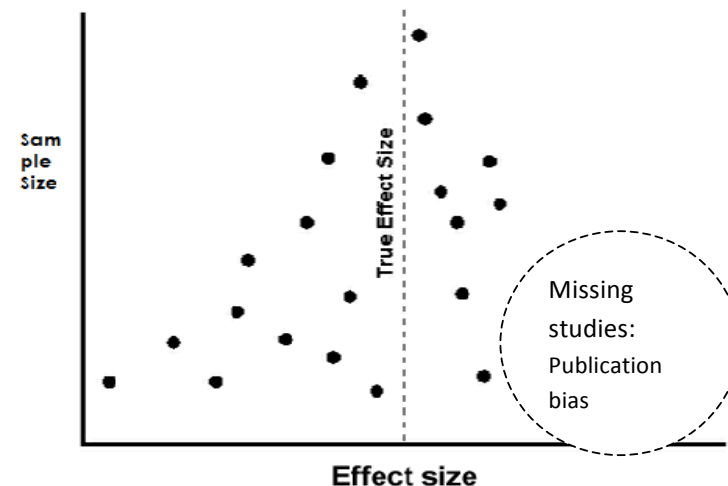
A measure of **effect size** (Cohen's d, OR or RR) is plotted on the x-axis

On the y-axis, the studies with the larger sample size (or, the lower SE) are plotted higher while those with, the smaller sample size (or greater SE) are plotted lower in the graph.

If a vertical line is drawn through the point representing the “true effect size” (obtained from the meta- analysis), we would expect the **larger studies** (the top end) to be closer to the line, and the **smaller studies** (those at, the lower end) to be more spread from the line of true effect size. This is because larger studies are more representative and have results closer to true effect size.

If the plot assumes an inverted funnel, as in the above figure, we can assume that there is no publication bias.

In the figure below, there is an obvious void on the right-hand corner of the plot – which represents the presence of a publication bias



ROC CURVE

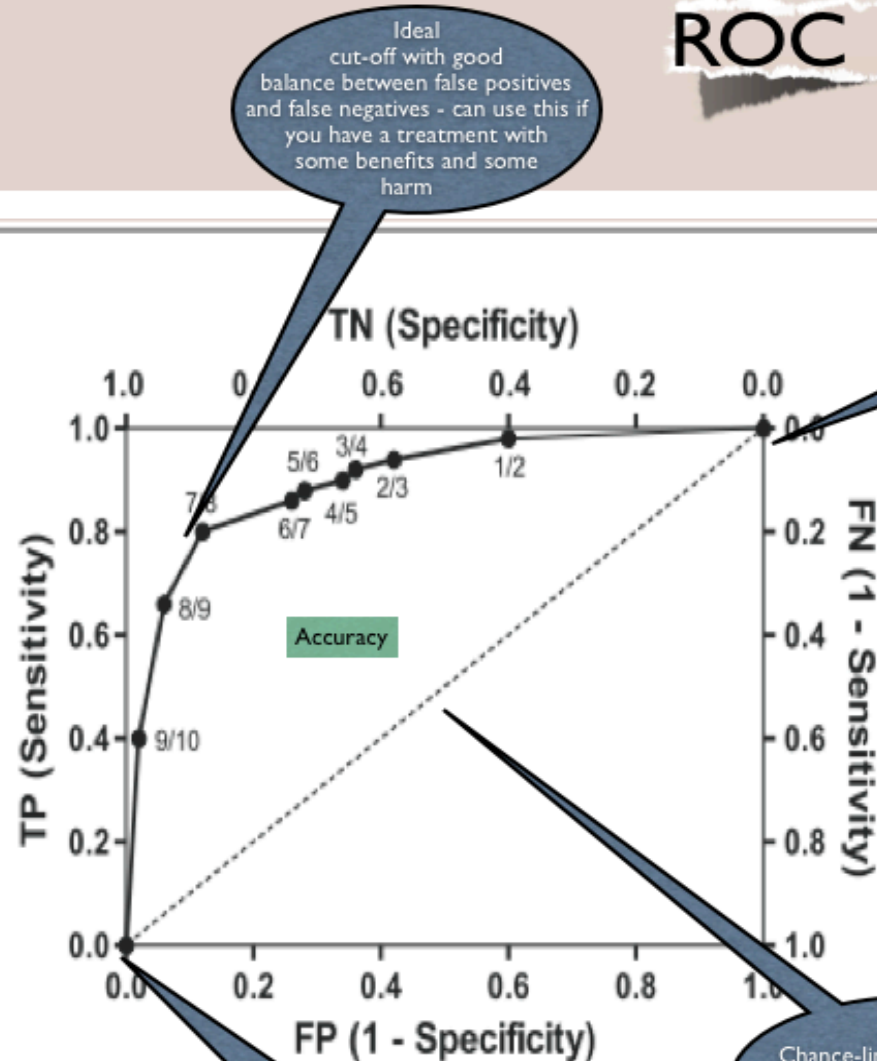
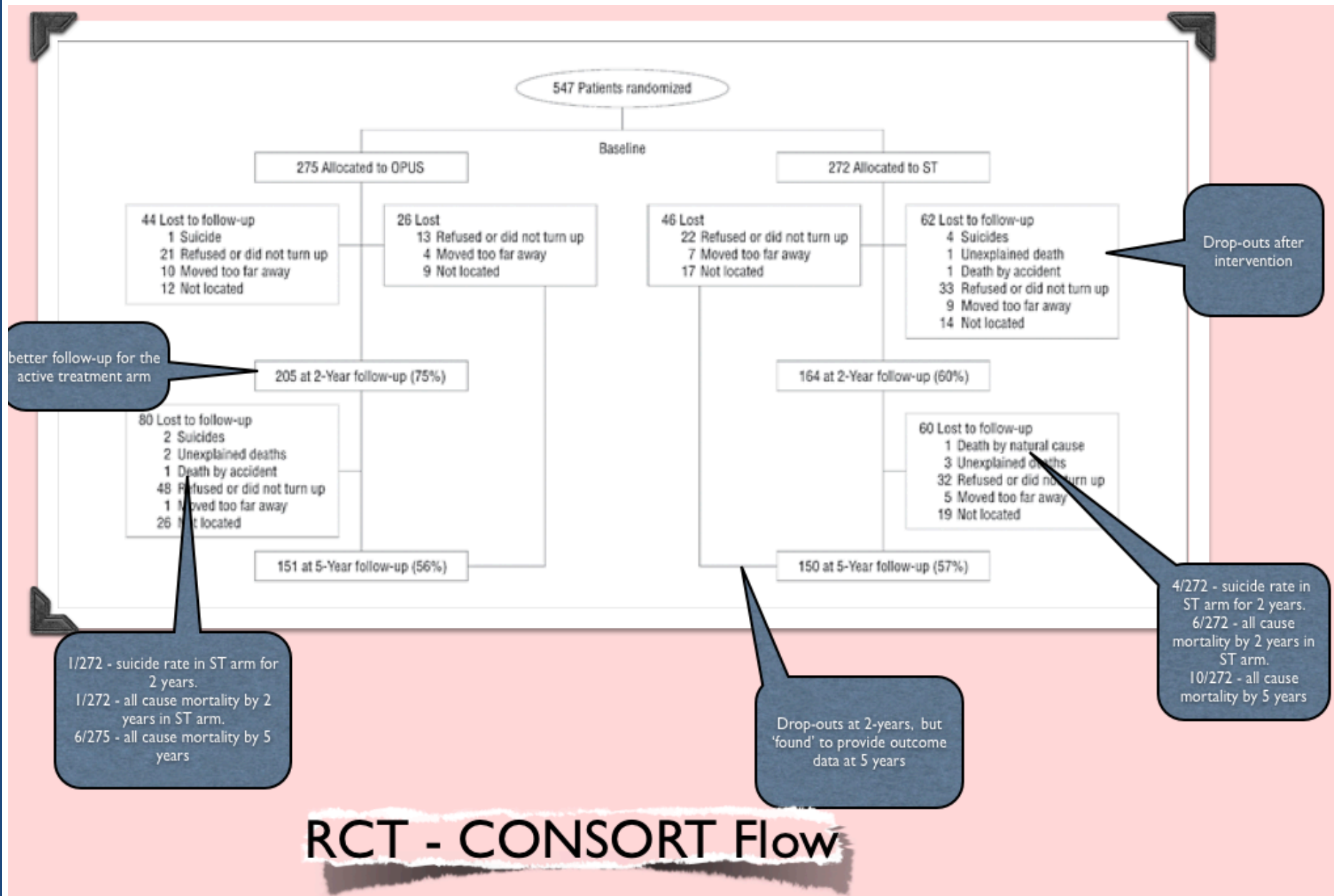


Table 3 Sensitivity and (1 - Specificity) for each cut point of the SPNP

Cut point	Sensitivity	1 - Specificity
< 1	1.00	1.00
1 / 2	0.98	0.60
2 / 3	0.94	0.42
3 / 4	0.92	0.36
4 / 5	0.90	0.34
5 / 6	0.88	0.28
6 / 7	0.86	0.26
7 / 8	0.80	0.12
8 / 9	0.66	0.06
9 / 10	0.40	0.02
> 10	0.00	0.00



RCT - CONSORT Flow

Forest Plot

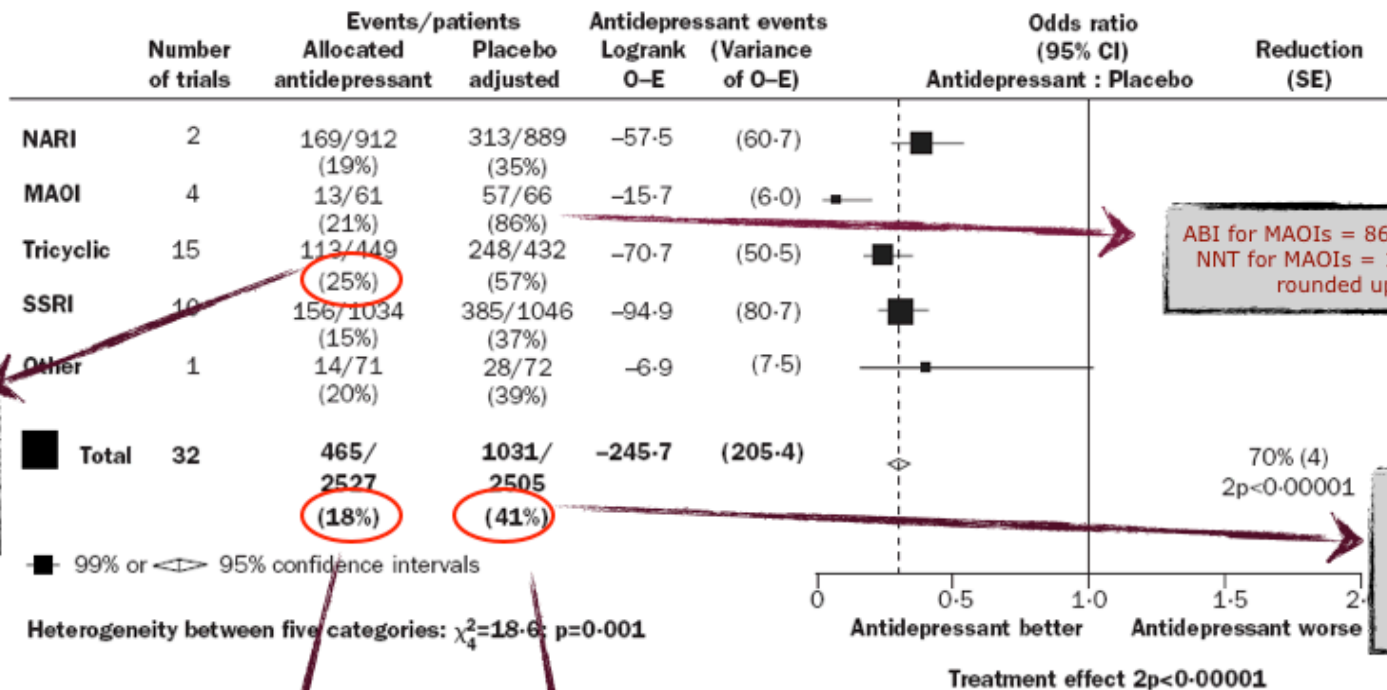


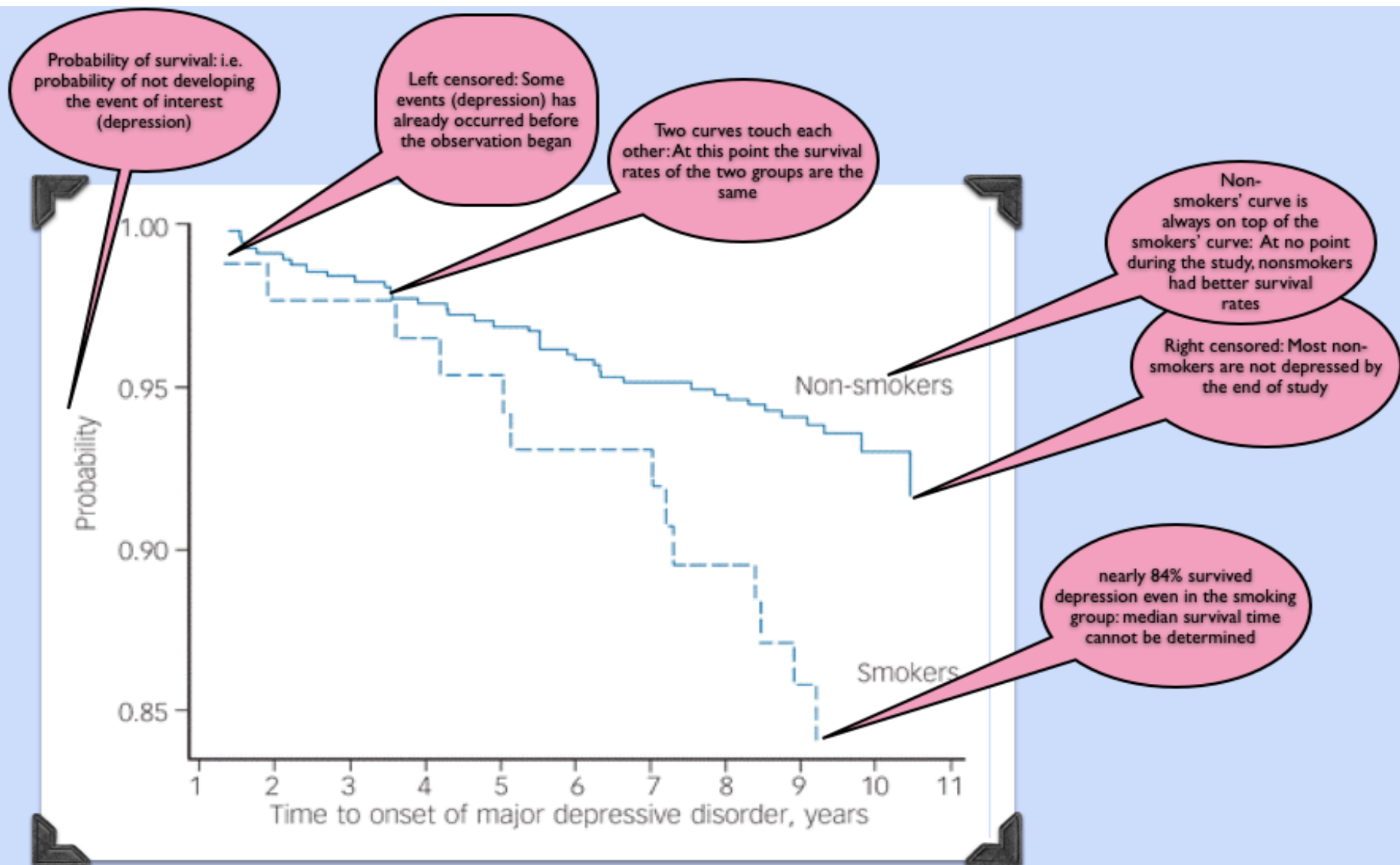
Figure 1: Risk of relapse: odds ratios by drug type

NARI=noradrenaline (norepinephrine) reuptake inhibitor. MAOI=monoamine oxidase inhibitor. SSRI=selective serotonin reuptake inhibitor.

The proportion that experienced a relapse while receiving TCAs (EER for TCAs)

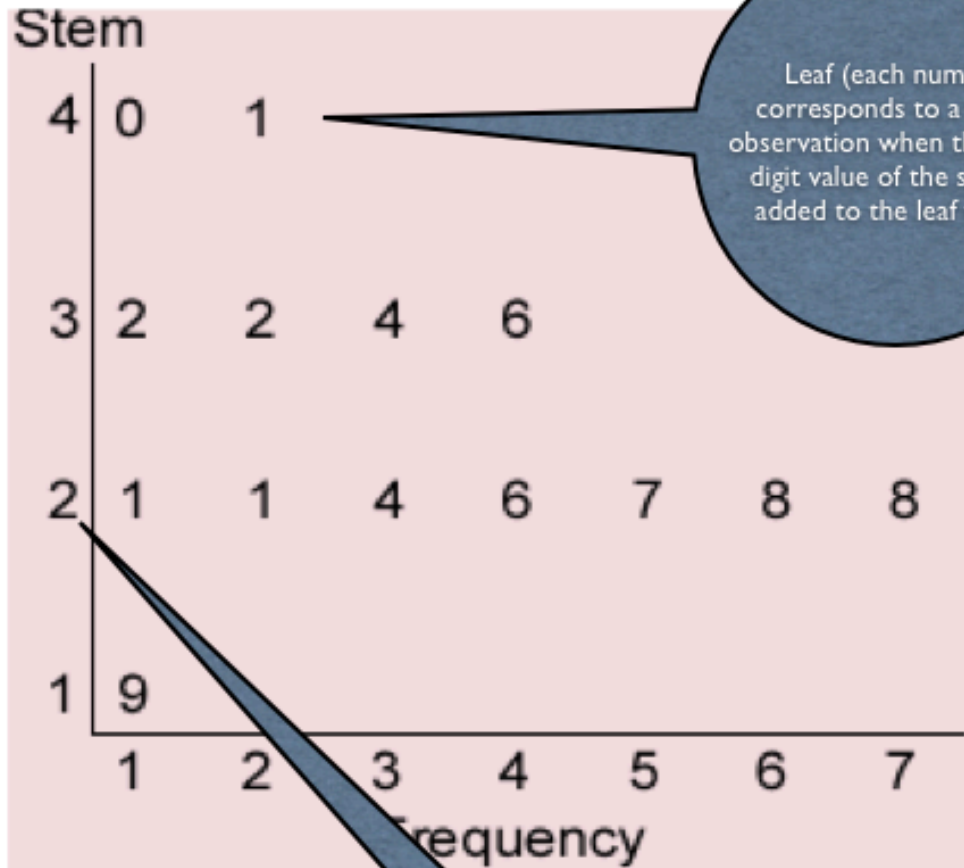
The proportion that experienced a relapse while receiving antidepressant: EER in pooled treatment group

The proportion that experienced a relapse while receiving placebo: the pooled control event rate



Kaplan Meier Plot

Stem and Leaf Plot



Sample:

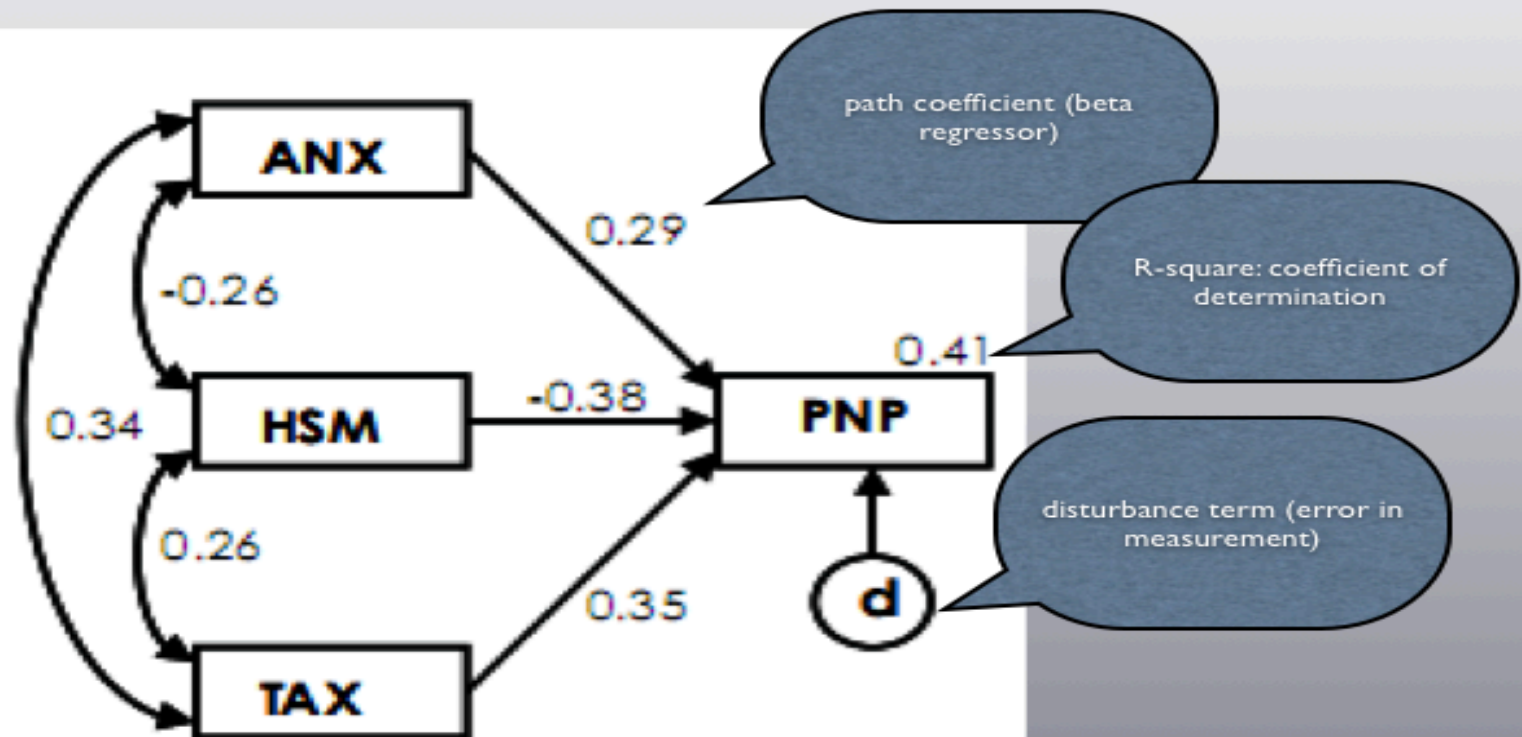
40,41

32,32,34,36

21,21,24,26,27,28,28

19

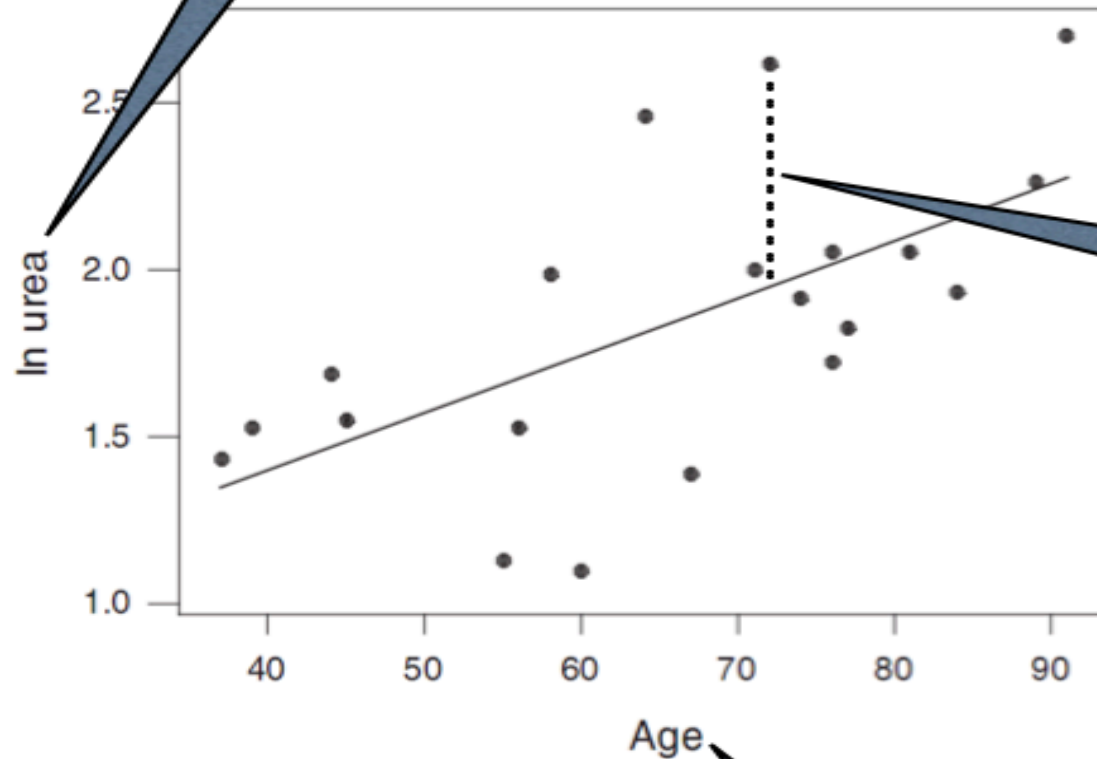
Path Analysis



All are endogenous variables in this model: all variables have arrows originating from and also pointing towards them. An exogenous variable will have no causative factor within the model

Scatter plot

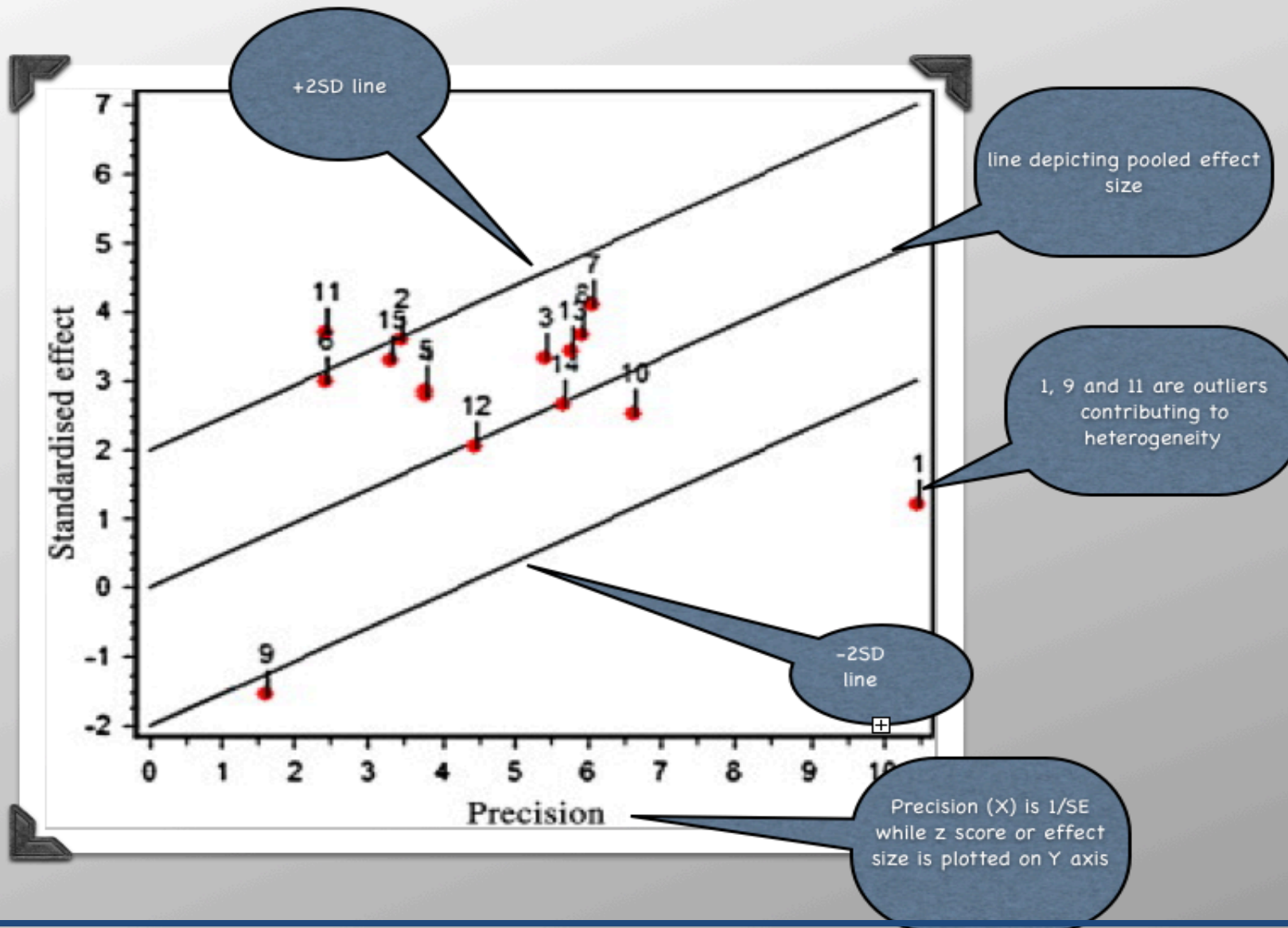
Log values may be used to yield parametric transformation



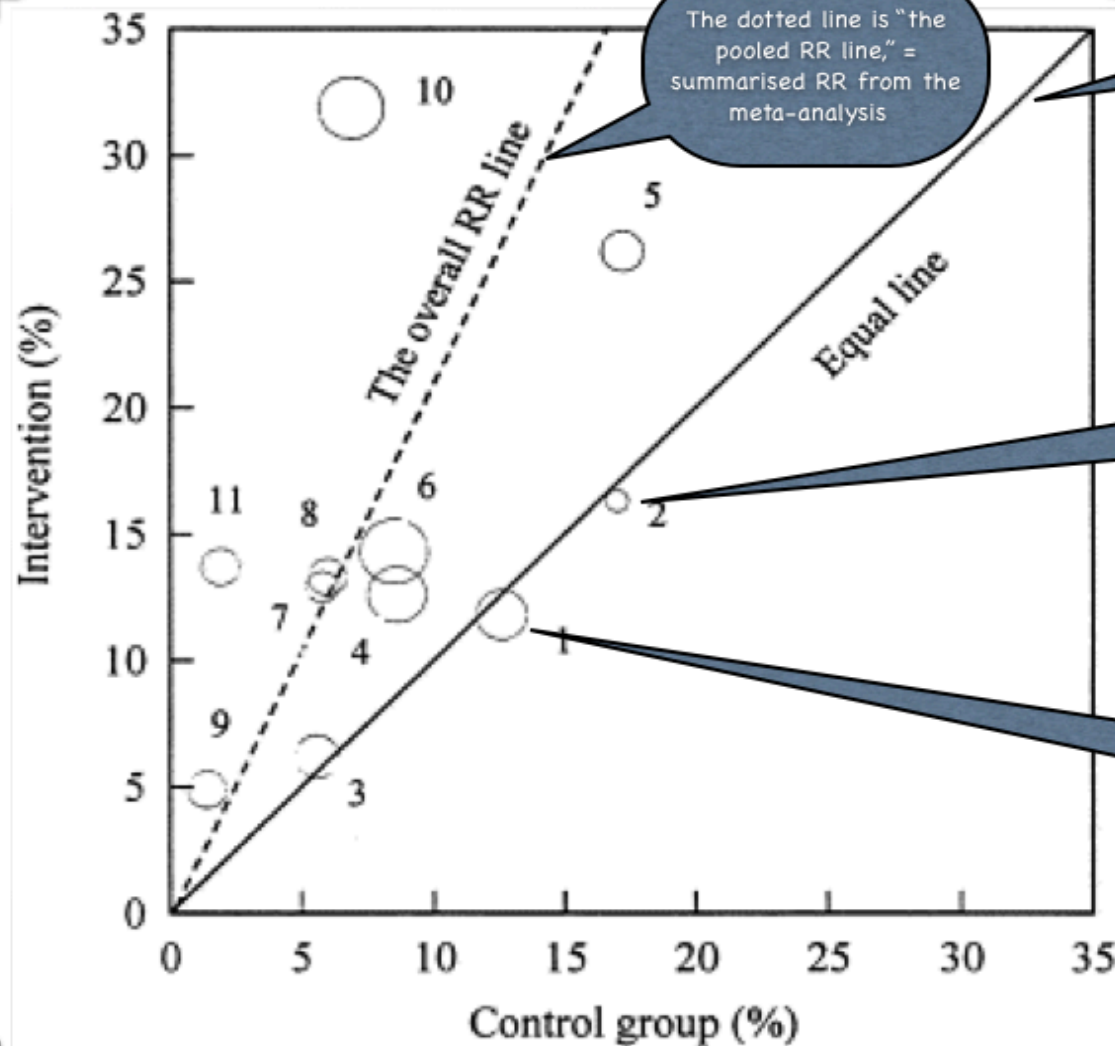
deviation (residual)

2 Continuous variables plotted against each other

Galbraith Plot



L'Abbe Plot



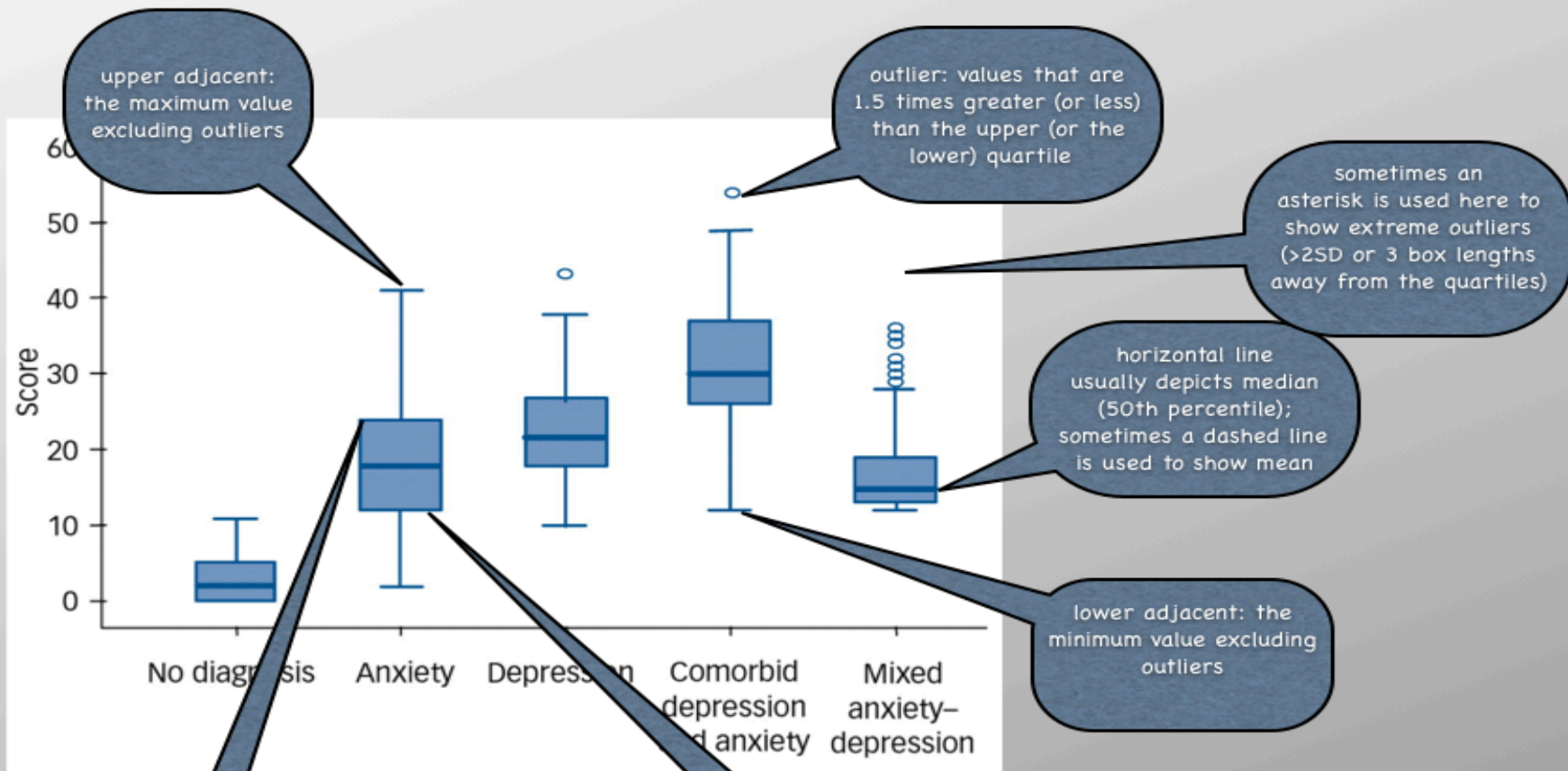
The dotted line is "the pooled RR line," = summarised RR from the meta-analysis

Equality line indicates that the event rates are equal in the two arms within trials ($RR = 1$)

Trials that are below the equality line show that the control groups have better event rates than the intervention groups

Each circle represents an individual trial; larger circles represent trials with more participants.

Box Whisker Plot



Das-Munshi, et al. The British Journal of Psychiatry 2008;192:171-177

MRCPsych Paper 3 Critical Review – Evidence-Based Practice Syllabic Content

Outcome: To make the optimal use of current best evidence in making decisions about the care of patients

1. Translation of clinical uncertainty into an answerable question

- 1.1. formulates clinical questions using the PECO(t) formula (Patient, exposure/intervention, comparison, outcome, time)
- 1.2. recognises and formulates different types of clinical questions:
 - 1.2.1. therapy
 - 1.2.2. harm
 - 1.2.3. aetiology
 - 1.2.4. prognosis
 - 1.2.5. diagnosis
 - 1.2.6. economic
 - 1.2.7. qualitative

2. Systematic retrieval of the best available evidence

- 2.1. Knows the different sources of evidence
- 2.2. Describes the “hierarchy of evidence” as it applies to different types of questions
- 2.3. Describes what is meant by:
 - 2.3.1. publication bias; and
 - 2.3.2. language of publication bias
- 2.4. Describes the difference between the following electronic databases:
 - 2.4.1. Cinahl
 - 2.4.2. Cochrane Library
 - 2.4.3. EMBASE
 - 2.4.4. PsycINFO
 - 2.4.5. Pubmed
 - 2.4.6. Sigle
- 2.5. Knows how research is catalogued and strategies for efficient retrieval
- 2.6. Searches efficiently and effectively:
 - 2.6.1. PubMed (Medline); and
 - 2.6.2. The Cochrane Library.